Operational Requirements

RE CH 2010-030 e

OPS Requirements

DAM tool improvement

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abstract Operational requirements for improvement of DAM tool in SKYVISU

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1 Endorsement

Decisional step 3 - APPROVAL	
Role	**Date/signature**
Raphaël MICHEL *skyguide* *OOR – Operational Change & Project Management*	
By signing this Operational Requirements, the **Change Owner** certifies that the planned change, with its needed actions included, can be conducted in conformance with that document.	
Decisional step 2 - VALIDATION	
Role	**Date/signature**
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Jérôme CORBOZ *skyguide* *OGDS – FMP Supervisor Geneva*	
Thomas DOSTER *skyguide* *OZC-T – DM Equipment ACC Zurich*	
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Peter NIEDERER *skyguide* *OZT-T – DM Equipment TWR/APP Zurich*	
Alexandre ZOULALIAN *skyguide* *OGTE – DM Equipment TWR/APP Geneva*	
By signing these Operational Requirements, the **Domain Manager Equipment** certifies that these Requirements match with the purpose of the related request(s) (refer to Chapter 4) and that the impact of these Requirements on his business unit has been coordinated within his unit (i.e. with DM Safety, Training & Equipment).	
Silvia MEYLAN *skyguide* *OORB – Operational Change & Project Management]*	Silvia MEYLAN Digitally signed by Silvia MEYLAN DN: cn=Silvia MEYLAN, o=skyguide, ou=OORB, email=silvia.meylan@skyguide.ch, c=CH Date: 2011.06.06 12:20:30 +02'00'
By signing these Operational Requirements the **Change Manager / Project Leader** certifies that all necessary actions have been completed to ensure that these requirements meet the needs of the change (refer to Chapter 4).	
Decisional step 1 - PREPARATION	
Role	**Date/signature**
Silvia MEYLAN *skyguide* *OORB – Operational Change & Project Management*	Silvia MEYLAN Digitally signed by Silvia MEYLAN DN: cn=Silvia MEYLAN, o=skyguide, ou=OORB, email=silvia.meylan@skyguide.ch, c=CH Date: 2011.06.06 12:20:15 +02'00'
By signing these Operational Requirements, the **Author** of document certifies that its content is complete and correct and that it was established based on application of official internal procedures.	

2 Overview

Many SIR have reported deficiencies with the DAM tool. During a SPVR seminary held in December 2008, a brainstorming of these deficiencies conducted to the initial CR2045 summarizing the main issues. Meetings and bilateral discussions were held with the Domain Managers (procedures, equipment and safety), SPVR, FMP operators and technical people in Zurich and in Geneva.

Concerning Zurich, no major problems are indicated because improvements were brought there approximately 3 years ago.

To consider that the Airspace CIV/MIL is differently managed in Zurich (MIL ON philosophy: the airspace belongs to military control and can be segregated from military to civil control) and in Geneva (MIL OFF philosophy: the airspace belongs to civil control and can be segregated from civil to military control). An implementation of harmonization of the philosophy was tried but unsuccessfully.

Concerning Geneva several problems are mentioned and discussed. These problems can be classified in two groups: a major problem for the FMP operators due to the complexity of the procedures and a major problem for the ATCO due to the complexity (overlapping) of the display of the information related to the available airspaces.

The collected problems have been listed, analyzed and classified according to the following categories: parameter change, evolution of current software, new development and procedure change. The result of this classification indicates that the majority of the problems can be solved by an evolution of the current tool and by an improvement of some procedures. One problem was already solved by the implementation of the requests 2157 and 2201 on October 22nd 2009, and some others are more related to the complexity of some procedures which mitigate the limitations of the DAM tool.

A basic analysis of several SIRs (367,432,640,642,643 and 649) shows that there are problems related to the display of multiple active zones and problems related to the display of the first available levels. The information given to the controllers can be incomplete, superfluous or ambiguous.

The goal is to improve DAM tool for ATCOs and FMP operators, increase efficiency, improve HMI. DAM tool has to analyze and define the main map that has to be displayed at the corresponding sectors without HMI ambiguity according:

- intelligent display = no superfluous or useless display
- management of the transparency in case of overlapping
- clear display of available/non-available airspace

Intermediate steps shall be provided for the prototyping.

Operational Deployment Date (ODD) is proposed on May 2012 and shall be defined in due time and confirmed by RP meeting

3 History

Version	Status	Date of issue	Description
0.1	Draft	2010-02-17	Creation of the document.
0.2	Draft	2010-05-05	Considered feedback from OORB (L. Bosonnet, S. Tabti) and OORC (T. Villiger) Update distribution list . The colour coding (green=new, yellow=modify and red=remove) is applied to differentiate the requirements according the current situation. Added requirement : HMI_029a, HMI_029b, HMI_034a, HMI_060a, HMI_065a to HMI_065b, HMI_069a to HMI_069g, HMI_071a to HMI_071e, HMI_073a to HMI_073g, HMI_074a, HMI_076a to HMI_076h, HMI_078a, HMI_078b, HMI_081a to HMI_081j, HMI_083a to HMI_083o, HMI_087a to HMI_087b, HMI_100a, HMI_101a, HMI_102a, HMI_112 to HMI_116, PAR_010 to PAR_028 Modified requirement : HMI_002, HMI_004, HMI_005, HMI_013 to HMI_014, HMI_018, HMI_022, HMI_026 to HMI_029, HMI_034, HMI_036, HMI_037, HMI_042, HMI_045 to HMI_046, HMI_052, HMI_060, HMI_065, HMI_067, HMI_069, HMI_071, HMI_073, HMI_074, HMI_076 to HMI_078, HMI_080 to HMI_083, HMI_087, HMI_095, HMI_098, HMI_102, HMI_103, HMI_106, HMI_108 to HMI_111, PAR_001, PAR_003 to PAR_007 Deleted requirement : HMI_053 to HMI_059, PAR_002, PAR_008, PAR_009
0.3	Draft	2010-06-24	Considered feedback from OGCE: wrong indication of sectors in the figures § 5.2.6.2. Considered feedback from OOR. Considered meetings held on the 17th of May in GVA & ZRH. Considered meeting (FMP operators) held on the 14th of June in GVA and feedback from A. Collins and T. Villiger
0.4	Draft	2010-07-08	Considered feedback from OORB (L. Bosonnet, S. Tabti) and OORC (T. Villiger); renumbered all the requirement.
0.5	Draft	2010-08-03	Considered feedback from TDA (Ch. Rey): - reorganized structure version A & B - added example of the display of an activated dynamic map - in the "Dynamic Maps Creation Window": added "Remove" button and modified requirements for "Add" button. Added requirement: HMI_355 to HMI_363 Modified requirement: HMI_004 to HMI_005, HMI_014, HMI_058, HMI_060, HMI_067, HMI_125, HMI_151 to HMI_154, HMI_165, HMI_169, HMI_191 to HMI_192, HMI_195, HMI_197, HMI_250, HMI_262, HMI_281, HMI_301, HMI_303, HMI_308, HMI_319, HMI_322, HMI_324, HMI_326, HMI_328, HMI_332 to HMI_333

			Deleted requirement: HMI_064 to HMI_065, HMI_068 to HMI_071, HMI_076 to HMI_124, HMI_126 to HMI_141, HMI_155 to HMI_163, HMI_166 to HMI_168, HMI_170 to HMI_190, HMI_244 to HMI_249, HMI_252 to HMI_261, HMI_264 to HMI_280, HMI_283 to HMI_287, HMI_300 to HMI_302, HMI_304 to HMI_307, HMI_309 to HMI_311, HMI_318, HMI_320 to HMI_321, HMI_323, HMI_325 to HMI_327
0.6	Draft	2011-04-14	Considered review made feedback from TVI: - remove the altitude correction "500ft" from the Dynamic map creation window of Zurich - remove all the requirements related to the lateral buffer - addition of a new value for the level correction sent by INCH (zrh only)
0.7	Proposed	2011-05-19	Merge of the version A and B to facilitate the reading of the document. Considered feedback RMI. Proposed version.
0.8	Proposed	2011-06-01	Considered feedback during the OPS Meetings held the 23rd May in Geneva and the 25th of May in Zurich: HMI-050 and HMI_054 (QNH corrections), HMI_036 (remove PRN position from sectors in Zurich), HMI_024 (date), HMI_057 (behaviours of DAM activity window), HMI_288 (missing of unit/sector button), HMI_151, HMI_199, HMI_200, HMI_253, HMI_301, HMI_303, HMI_308, HMI_391, HMI_322, HMI_324, HMI_326, HMI_328 Requirements are discussed with TDA at CH level during a video conference meeting held the 31st May and feedback are considered: HMI_361 (TSA activation), HMI_354 (removed)
1.0	Released	2011-06-06	Released version

4 References

Ref	Document Name	Date
[1]	CR2045 – DAM OPS compliant CM : Silvia Meylan; CO : Raphaël Michel	2008-12-12
[2]	Operational Study - DAM tool v1.0 2010-02-23	2010-02-23

5 Requirements

Remark: only the highlighted requirements are relevant. The highlighting is done as following:

Added requirement
Modified requirement
Deleted requirement

The requirements that are not highlighted describe the status quo of the equipment and are not in the scope of the review.

5.1 General Requirements

RE 2010-030E GEN_001 Only the login of the FMP operator and of the SPVR shall allow the activation of the Dynamic Airspace Management (DAM) window.

RE 2010-030E GEN_003 FMP and SPVR shall be able to select predefined maps and complete them with all the data according to the activity.

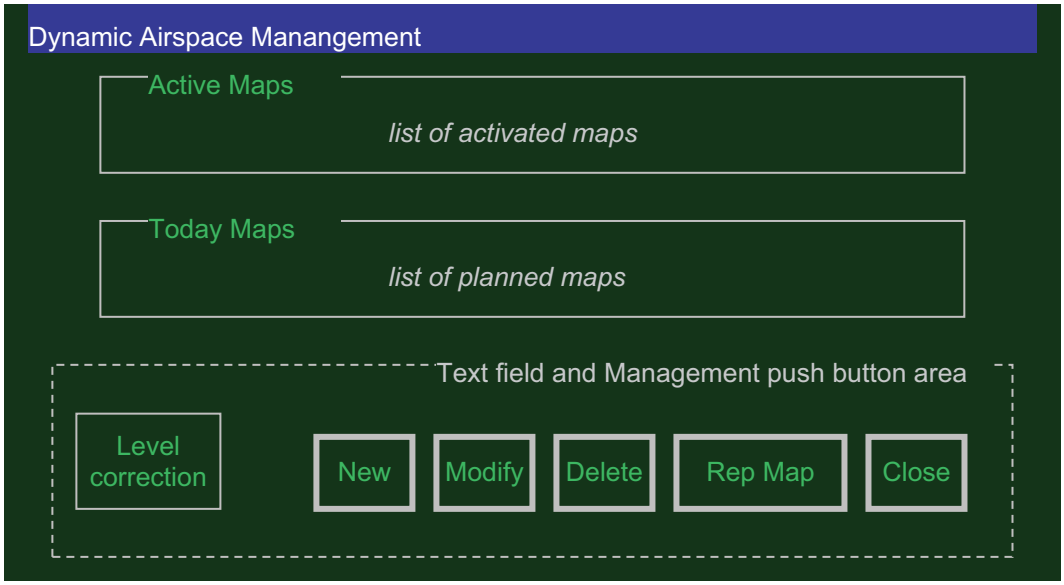
5.2 HMI Requirements (version A)

5.2.1 Dynamic Airspace Management (DAM) window

RE 2010-030E HMI_001 The DAM window shall be opened with a click B1 on the "DYNAMIC AIRSPACE" menu item of the Technical Setup Bar.

RE 2010-030E HMI_002 The DAM window shall be divided into three parts:

- **Active Maps** – Display area which contains the list of activated maps
- **Today Maps** – Display area which contains the list of maps that are planned to be activated on the current day or are terminated (end time has **been reached)**
- **Text field and management push button area** – which contains push buttons (New, Modify, Delete, Rep. Map and Close) and a text field indicating level correction provided by the INCH system.



RE 2010-030E HMI_003 For map entries within the "Active Maps" or "Today Maps" the following information shall be available:

- Active period for the current day: start time and end time
- Altitude range: lower and upper levels
- "ft" if the altitude are set to displayed in feet
- Map name
- For maps with a specific status (TSA), the activation period and altitude range are replaced by an indication of the status:
  Terminated, Coordination need

RE 2010-030E HMI_004 The line of text related to the "Active Maps" or to the "Today Maps" shall be displayed entirely up to the frame of the available space in the corresponding window (see also RE 2010-030E HMI_332).

ⓘ Shall not be truncated as it is currently the case

ⓘ The size of the window shall not change

RE 2010-030E HMI_005 If the text related to the "Active Maps" or to the "Today Maps" exceeds the frame of text delimitation (window) a hook on the line shall permit to display the text entirely (see also RE 2010-030E HMI_333).

5.2.1.1 Text field and management push button area

RE 2010-030E HMI_006 The "Level correction" text field on the "Dynamic Airpsace Manangement window" shall display the level sent by INCH:

- Ex. in GVA:
  first line "FL100:" and the FL correction to be applied to lower and upper map levels below 10000 feet
  second line: "FL180:" and the FL correction to be applied to lower and upper map levels above 10100 feet.

- Ex. in ZRH:
  first line "Min A9"
  second line: "FL140" or "FL150 or "FL160" that can be switched; a click B1 on the correction text field opens a confirmation window

RE 2010-030E HMI_007 The "New" push button shall be available for FMP and SPVR users only.

RE 2010-030E HMI_008 The "Modify" push button shall be available for FMP and SPVR users only.

RE 2010-030E HMI_009 The "Delete" push button shall be available for FMP and SPVR users only.

RE 2010-030E HMI_010 The "Rep. Map" push button shall be available for FMP and SPVR users only.

RE 2010-030E HMI_011 The "Close" push button shall be available for all users.

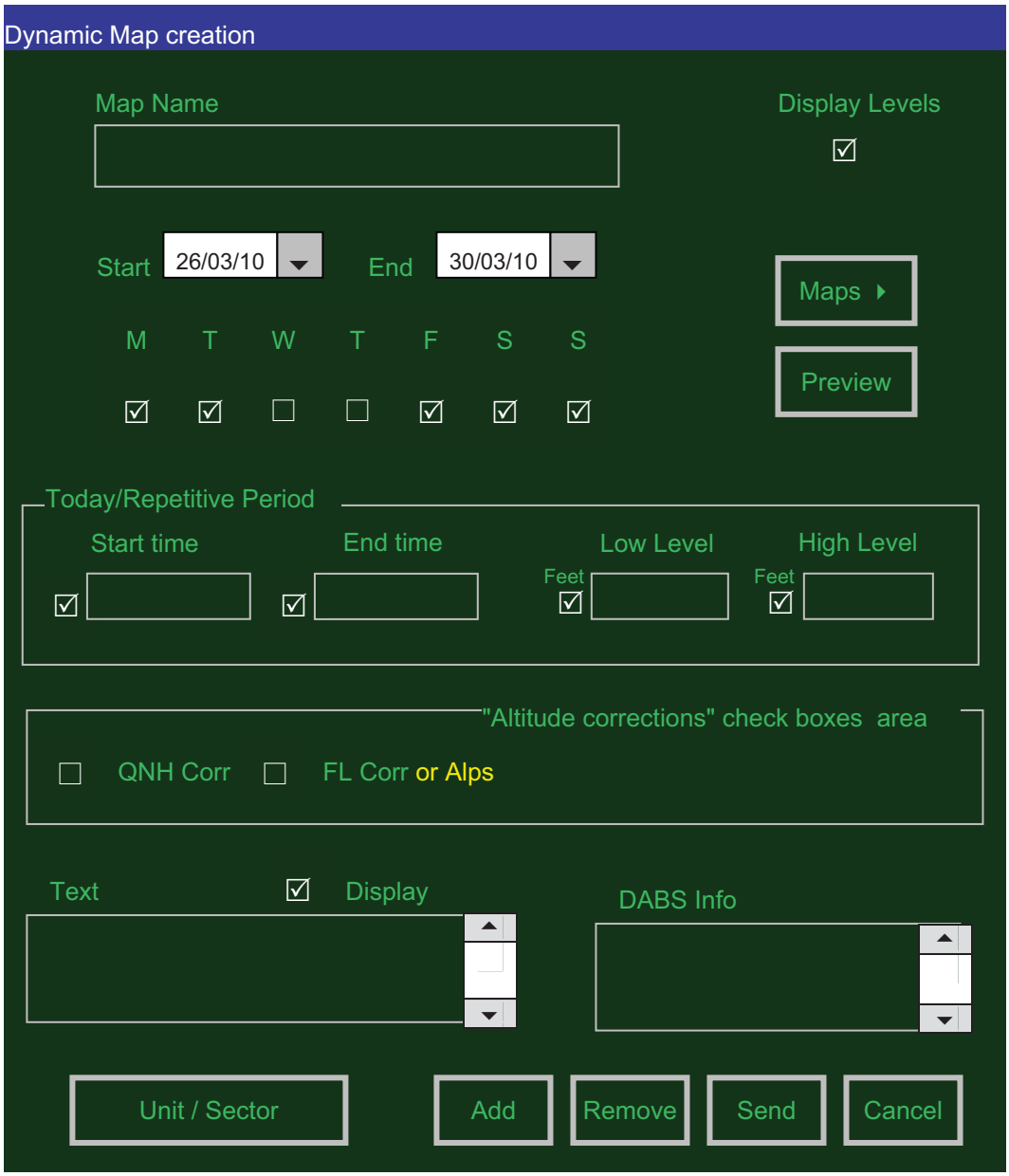
RE 2010-030E HMI_012 A Click B1 on the "Close" push button shall close the "Dynamic Airspace Management" window.

5.2.2 Dynamic Maps Creation Window

RE 2010-030E HMI_013 A Click B1 on the "New" push button shall open the "Dynamic Map Creation" window.

RE 2010-030E HMI_014 The "Dynamic Maps Creation" window shall contain the following elements:

- "Map Name" edit text field – see details in § 5.2.2.1
- "Start" and "End" combo boxes (date) – see § 5.2.2.2
- "Today/Repetitive periods" – see details in § 5.2.2.3
- "Display level" check box – see details in § 5.2.2.5
- "Altitude corrections" check boxes – see details in § 5.2.2.6
- "Multi-line" text fields – see details in § 5.2.2.7
- Push button "Unit / Sector" – see details in § 5.2.2.4
- Push buttons – see details in § 5.2.2.8



5.2.2.1 "Map Name" edit text field

RE 2010-030E HMI_015 The "Map Name" edit text field shall allow to enter the name of the created dynamic map.

5.2.2.2 "Start" and "End" combo boxes (date)

RE 2010-030E HMI_016 The "Start" and "End" combo boxes shall allow to select the active period of the created dynamic map.

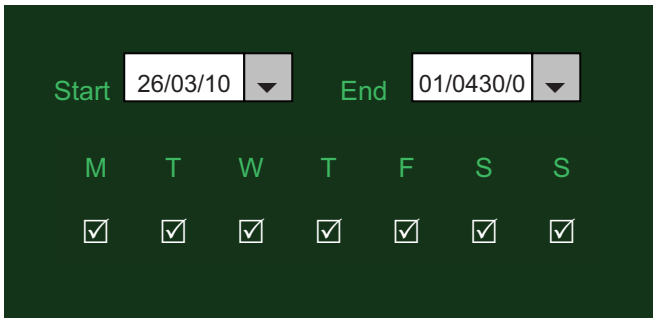
RE 2010-030E HMI_017 By default "Start" and "End" days shall be set to the current

date.

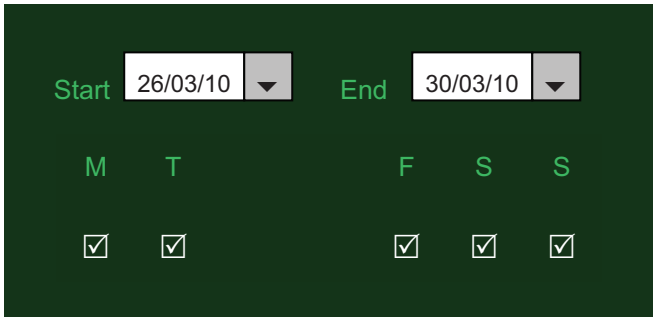
RE 2010-030E HMI_018 Click B1 on the arrow push button of the combo boxes shall open a calendar window which allows by month to select the date.

RE 2010-030E HMI_019 If the "End" date is different from the "Start" date, 7 check boxes (one per day of the week) shall automatically appear below "Start" and "End" combo boxes.

ⓘ "End" date shall be defined beyond the "Start" date



RE 2010-030E HMI_020 By default only the check boxes of the possible corresponding day included in the selected period shall be ticked and displayed (see figure below).



RE 2010-030E HMI_021 It shall be possible to de-select a day included in the period.

RE 2010-030E HMI_022 If one check box is un-ticked, the repetitive map shall not be active for the corresponding day during all the selected period.

RE 2010-030E HMI_023 It shall be possible to modify at any time the "Start" date and the "End" date and the check boxes shall be automatically updated according to the new dates.

RE 2010-030E HMI_024 In case of a modification of the "Start" date, the "End" date shall be updated automatically accordingly and shall keep the constant duration.

5.2.2.3 "Today/Repetitive periods"

RE 2010-030E HMI_025 The "Today/Repetitive Period" allows to define the daily activation information.

RE 2010-030E HMI_026 For the "Today/Repetitive Period" the following information

can be set:

- Check box for Start indication
- "Start" time
- Check box for End indication
- "End" time
- Check box for Low level indication in feet
- "Low" level
- Check box for High level indication in feet
- "High" level

ⓘ Check box for Start indication is not available for TSA in ZRH

RE 2010-030E HMI_027 The "Today/Repetitive Periods" shall always be defined on the same day.

RE 2010-030E HMI_028 When the "Start time" check box is ticked, a Start indication shall be displayed 15 minutes before the map activation at the CWP.

RE 2010-030E HMI_029 When the "End time" check box is ticked, an End indication shall be displayed 15 minutes before the map de-activation at the CWP.

RE 2010-030E HMI_030 When the focus is in the "Start time" text field pressing the key "N" (standing for Now) of the keyboard shall fill the text field with the current system time and shall set the focus in the "End" text field.

RE 2010-030E HMI_031 When the focus is in the "End time" text field pressing the key "E" (standing for End) of the keyboard shall fill the text field with the last possible time of the day: "23:59".

RE 2010-030E HMI_032 When a predefined radar map is selected, by default the proposed value into "Low level" text field and "High level" text field shall correspond to the parameters of the predefined map.

ⓘ see more 5.3.1

RE 2010-030E HMI_033 If no parameters are defined in the predefined map and no input is performed into "Low level" text field and "High level" text field (Altitude range), the following default values in hundreds of feet shall be applied:

- "000" and "630" in GVA
- "000" and "999" in ZRH

RE 2010-030E HMI_034

When the check box "Low level indication in feet" is ticked the unit for the low level during the period shall be displayed in feet.

ⓘ see more 5.3.1

RE 2010-030E HMI_035

When the check box "High level indication in feet" is ticked the unit for the high level during the period shall be displayed in feet.

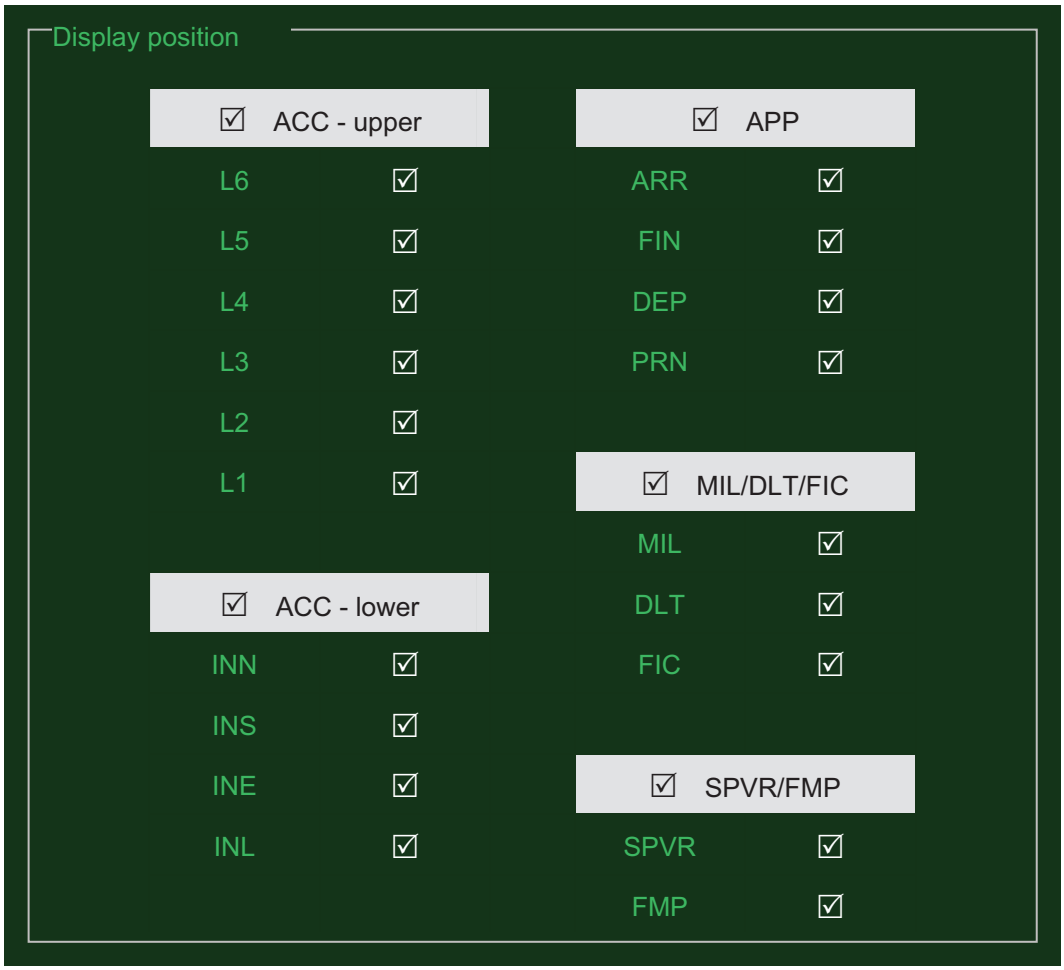
ⓘ see more 5.3.1

5.2.2.4 "Unit / Sector" push button

RE 2010-030E HMI_036

The push button "Unit / Sector" shall open the "Display Position" window that allow to select the units and/or the sectors where the map shall be displayed when activated.

In Geneva:



In Zurich:

RE 2010-030E HMI_037 By default all the "unit" check box and the "sector" check box shall be ticked as defined in the predefined map.

ⓘ see more § 5.3.4

RE 2010-030E HMI_038 It shall be possible to tick check boxes of a "unit" and/or a "sector" that is un-ticked.

RE 2010-030E HMI_039 It shall be possible to un-tick check boxes of a "unit" and/or a "sector" that is ticked.

RE 2010-030E HMI_040 If all the "sector" check boxes of a "Unit" are ticked, the "unit" check box shall automatically be ticked.

RE 2010-030E HMI_041 If a "unit" check box is ticked all the related "sector" check boxes shall be ticked.

RE 2010-030E HMI_042 If the last "sector" check box is manually un-ticked, the related "unit" check box shall be un-ticked automatically.

RE 2010-030E HMI_043 If a "unit" check box is manually un-ticked, all the related "sector" check boxes shall be un-ticked automatically.

RE 2010-030E HMI_044 At least a sector shall be ticked, otherwise an error message shall be displayed.

5.2.2.5 "Display levels" check box

RE 2010-030E HMI_045 The "Display levels" check box shall be ticked/un-ticked by default as defined in the predefined map.

ⓘ See more § 5.3.2

RE 2010-030E HMI_046 The status of the "Display levels" check box shall affect the map level indications as following:

- If ticked the level indication shall be displayed at the CWP (see more § 5.2.9.1)
- If un-ticked, the level indication shall not be displayed

5.2.2.6 "Altitude corrections" check boxes

RE 2010-030E HMI_047 The "FL Corr" check box and the "Alps" check box shall be ticked/un-ticked by default as defined In the predefined map.

ⓘ see more § 5.3.5

RE 2010-030E HMI_048 If the "FL Corr" check box is ticked the FL correction provided by the INCH system and displayed in the DAM window correction text field shall be applied for the map lower and upper levels as following:

- For levels under or equal to 10'000 ft, the correction FL100 is added.
- For levels under or equal to 10'000 ft, the correction FL110 is added.

RE 2010-030E HMI_052 If the "Alps" check box is ticked it shall increase the map upper level of 10 hundred feet in case of the display of "Min A9 FL150" or of "Min A9 FL160" in the DAM window correction text field.

RE 2010-030E HMI_049 If the "FL Corr" check box or the "Alps" check box is not ticked no FL correction shall be applied for the map.

RE 2010-030E HMI_050 If the "QNH Corr" check box is ticked it shall affect the upper levels of the displayed maps as following:

QNH	First usable level correction
QNH ≥ 1013	none
980 ≤ QNH < 1013	+ 1000 ft
QNH < 980	+ 2000 ft

RE 2010-030E HMI_054 It shall be possible to select the two corrections "QNH corr" and "Alps" simultaneously or the two corrections "QNH corr" and "FL corr".

RE 2010-030E HMI_053 The "500 ft" check box shall be removed from the "Altitude corrections".

5.2.2.7 "Multi-line" text fields and "Display" check box

RE 2010-030E HMI_055 The "Text" multi-line editable text field shall allow to provide additional comments to the map.

RE 2010-030E HMI_056 The status of the "Display text" check box shall affect the text indications as following:

- If "Display" check box is ticked the "Text" is displayed in a DAM activity window at all the CWP during the period of activity.
- If "Display" check box is un-ticked the "Text" is displayed in the Map information window (click B3).

RE 2010-030E HMI_057 A "DAM activity" window is a text window that provide information about all the DAM activities ticked by the operator as described in RE 2010-030E HMI_056.

It shall appear in an iconified status or displaying one or more lines of information at the CWPs (same behaviours applied to the MV entry window displayed in skyvisu).

RE 2010-030E HMI_058 The "text" set on the "Multi-line" text field of all the active maps that have the "Display" check box ticked shall be displayed in the unique "DAM activity" window according the time of activity.

RE 2010-030E HMI_059 The input text in the "Text" multi-line editable text field shall be limited to 250 characters.

RE 2010-030E HMI_060 The "DABS Info" multi-line non-editable text field shall summarize the geographic parameters for the DABS created dynamic maps.

5.2.2.8 Push buttons

5.2.2.8.1 "Maps"

RE 2010-030E HMI_125 A click B1 on the "Maps" push button shall allow to open a cascading submenu with the following:

- AIRSPACE MAPS
- AIRSPACE MAPS GROUP

- DABS (create a free map)

RE 2010-030E HMI_062 The cascading submenu of AIRSPACE MAPS shall allow to select a predefined map.

ⓘ see more § 5.2.3

RE 2010-030E HMI_156 The cascading submenu of AIRSPACE MAPS GROUPS shall allow to select several predefined map.

ⓘ see more § 5.2.4

RE 2010-030E HMI_063 The cascading submenu of DABS shall allow to create a free map.

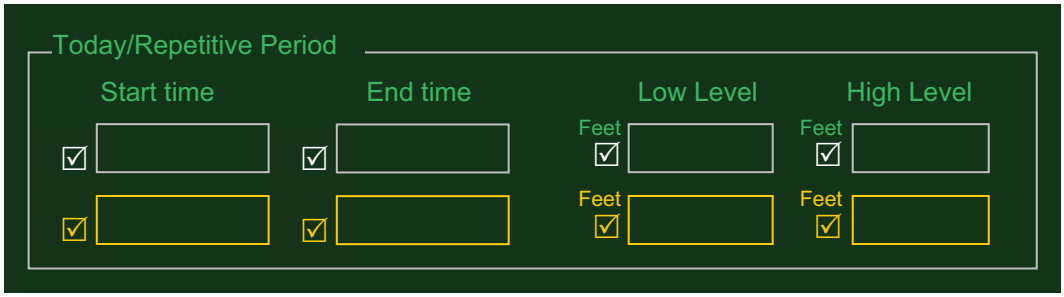
ⓘ see more § 5.2.5

5.2.2.8.2 "Preview"

RE 2010-030E HMI_066 A click B1 on the "Preview" push button shall display temporarily the selected map.

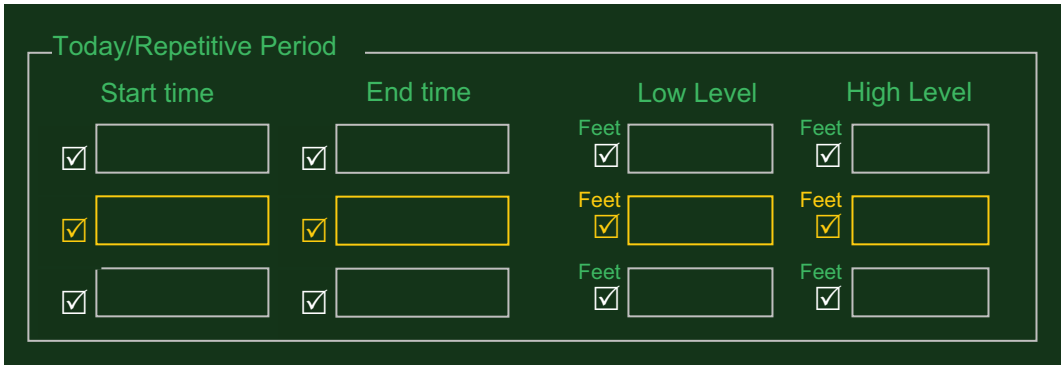
5.2.2.8.3 "Add"

RE 2010-030E HMI_067 A click B1 on the "Add" push button shall add a new row below the current one in the "Today/repetitive Period" part of the "Dynamic Airspace Creation" window to permit to set an additional period (see as example the figure below, the row in orange is added).



5.2.2.8.4 "Remove"

RE 2010-030E HMI_355 When the focus is in a field of a row in the "Today/repetitive Period" part of the "Dynamic Airspace Creation" window a click B1 on the "Remove" push button shall remove the entire row (see as example the figure below, the row in orange (the selected one) is removed).



5.2.2.8.5 "Send"

RE 2010-030E HMI_072 A click B1 on the "Send" push button shall close the "Dynamic Airspace Creation" window.

RE 2010-030E HMI_356 When the "Dynamic Airspace Creation" window is closed the inputs for the predefined map shall be validated and shall be sent to SKYVISU server.

RE 2010-030E HMI_073 The SKYVISU server shall distribute the new dynamic map to all the corresponding CWPs.

RE 2010-030E HMI_074 If the created map is set to be activated on the current day an entry is added in the "Today Maps display" area in the "Dynamic Airspace management" window (RE 2010-030E HMI_002).

5.2.2.8.6 "Cancel"

RE 2010-030E HMI_075 A click B1 on the "Cancel" push button or a click B2 on the title bar of the window shall cancel the current creation settings (inputs are lost) and shall close the "Dynamic Map Creation" window.

5.2.3 Dynamic Maps Creation – "AIRSPACE MAPS" cascading submenu

RE 2010-030E HMI_142 The "AIRSPACE MAPS" cascading submenu of "Maps" push button shall allow to select a predefined map available in the menu.

RE 2010-030E HMI_143 When the predefined map is selected the predefined map name shall be displayed in the "Map name" edit text field in the "Dynamic Map creation" window.

RE 2010-030E HMI_144 When the predefined map is selected by default "Start day" and "End day" combo boxes in the "Dynamic Map creation" window shall be set to the current date.

RE 2010-030E HMI_145 When the predefined map is selected by default the edit text fields and check boxes of the "Altitude range" in the "Dynamic Map creation" window shall be set according the parameters defined in the predefined map.

ⓘ see more § 5.3.1

RE 2010-030E HMI_146

When the predefined map is selected by default the "Display levels" check box in the "Dynamic Map creation" window shall be updated according the parameters defined in the predefined map.

ⓘ See more § 5.3.2)

RE 2010-030E HMI_147

When the predefined map is selected by default the Unit/Sector check boxes in the "Display positions" window shall be updated according the parameters defined in the predefined map.

ⓘ see more § 5.3.4

RE 2010-030E HMI_148

When the predefined map is selected by default the "Altitude corrections" check boxes in the "Dynamic Map creation" window shall be updated according the parameters defined in the predefined map.

ⓘ see more § 5.3.5

RE 2010-030E HMI_149

When the FMP operator has selected the predefined map, he shall be allowed to set differently the following proposed elements in the "Dynamic Map creation" window:

- Map start and ends days
- Map active days
- Map activation periods (start and end times)
- Map altitude range (Low / High level and feet)
- Map altitude correction
- Map text comments
- Map display at unit and sector

5.2.4 Dynamic Maps Creation – "AIRSPACE MAPS GROUP" cascading submenu

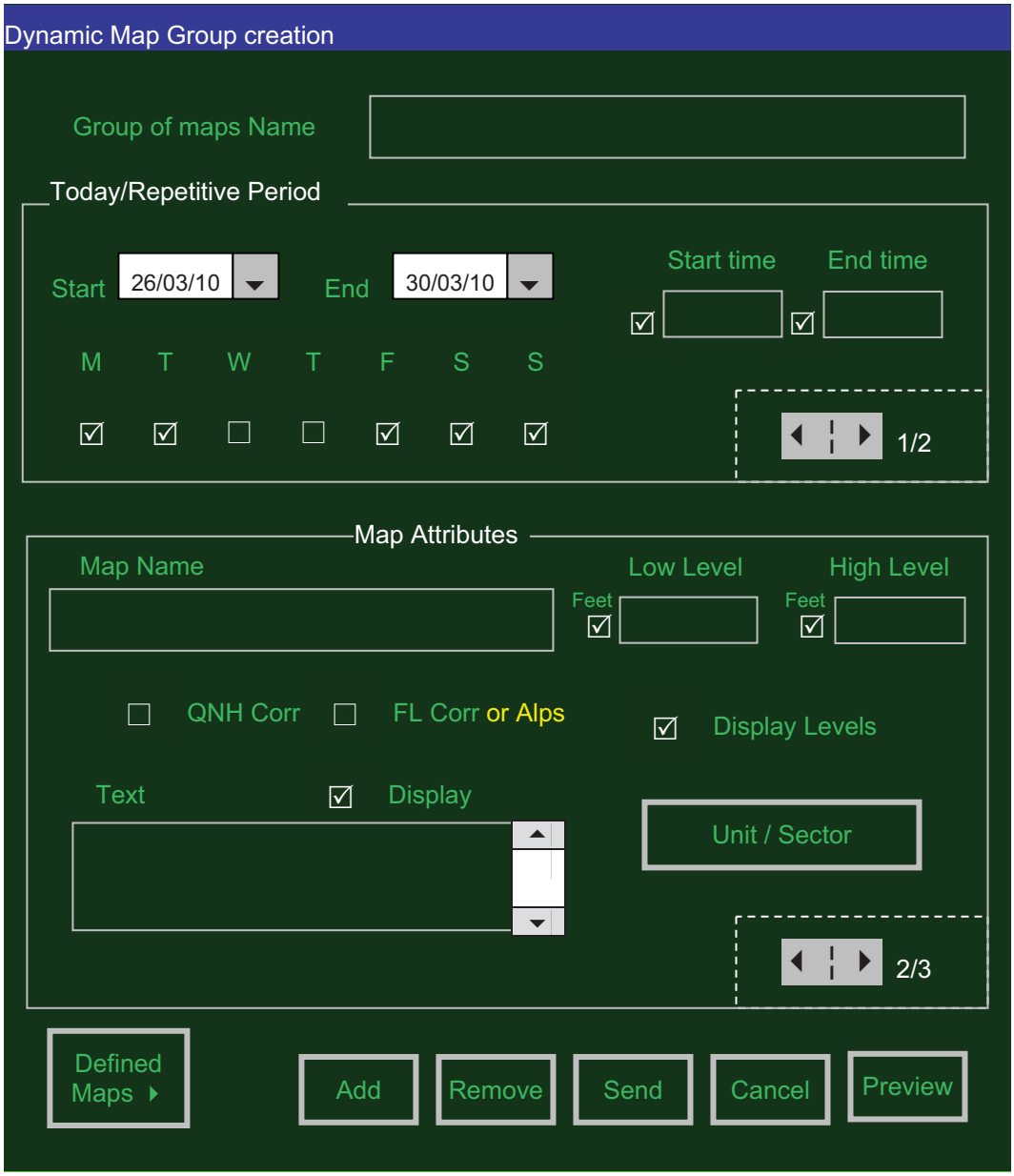
RE 2010-030E HMI_150

The "AIRSPACE MAPS GROUP" cascading submenu of "Maps" push button shall open the "Dynamic Map Group Creation" window.

RE 2010-030E HMI_151

The "Dynamic Maps Group Creation" window shall contain the following elements:

- Group of maps Name
- "Today/Repetitive periods" – see details in § 5.2.4.1
- "Map attributes" – see details in § 5.2.4.3
- Push buttons – see details in § 5.2.4.4



5.2.4.1 "Group of maps Name"

RE 2010-030E HMI_253 The "Group of maps Name" area allows to define the name of the group that is composed of all the selected maps.

5.2.4.2 "Today/Repetitive periods"

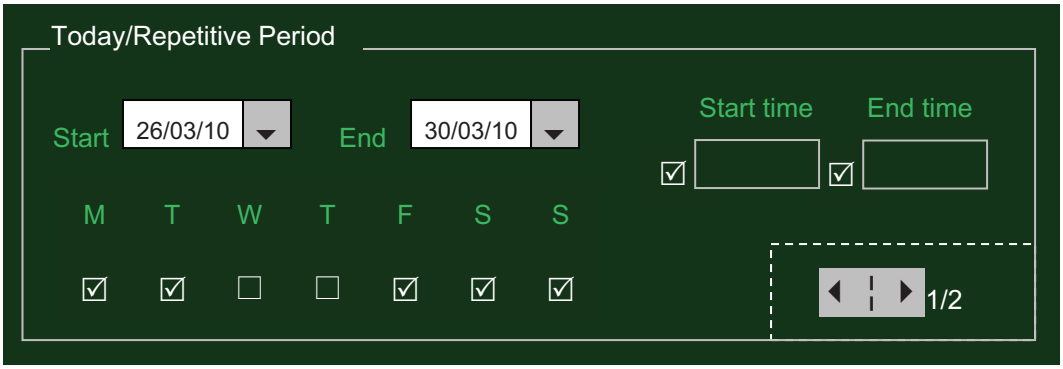
RE 2010-030E HMI_152 The "Today/Repetitive Period" area allows to define the daily activation information for a group of maps.

RE 2010-030E HMI_153 In the "Today/Repetitive Period" area the following information can be set (see figure below):

- Combo box for Start date
- Combo box for End date
- Check box for Day in the week
- Check box for Start indication

- "Start" time
- Check box for End indication
- "End" time
- "Preview /Next" page

ⓘ Check box for Start indication is not available for TSA in ZRH



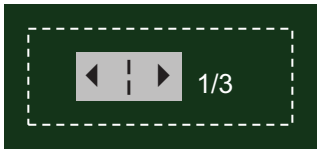
RE 2010-030E HMI_154 For the "Start" and "End" combo boxes the same features described in § 5.2.2.2 shall be applied (from RE 2010-030E HMI_017 to RE 2010-030E HMI_024).

RE 2010-030E HMI_164 The "Today/Repetitive Periods" shall be defined for all the selected dynamic map of the group.

RE 2010-030E HMI_165 For the "Start time" check box and the "End time" check boxes the features described in § 5.2.2.3 shall be applied (from RE 2010-030E HMI_028 to RE 2010-030E HMI_031).

RE 2010-030E HMI_358 To permit to display each "Today/Repetitive Period" area created when more than one period map is selected the following shall be displayed (see figure below):

- "Preview/Next" arrows (to move from a "Map attribute" area to the other)
- a counter (to indicate how much maps are composing the group)



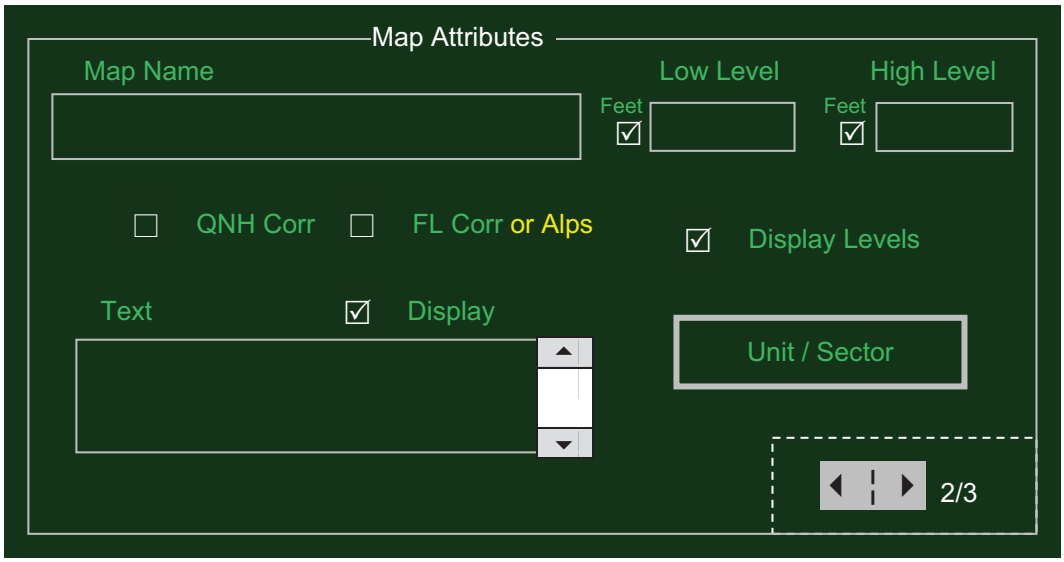
5.2.4.3 "Map attributes"

RE 2010-030E HMI_359 The "Map Attributes" area allows to define the attributes for each selected map of the group of maps.

RE 2010-030E HMI_360 In the "Map Attributes" area the following information can be set (see figures below):

- Map Name
- Check boxes for feet (Low and High level)
- Low and High level

- Check boxes for Altitude corrections
- Text multi-line field and related check box
- Unit/Sector
- "Preview /Next" page



RE 2010-030E HMI_169 For the fields in the "Map Attributes" area the features described in the following requirements from § 5.2.2.1, § 5.2.2.3 (HMI_032 to HMI_035 only), 5.2.2.4, 5.2.2.5, 5.2.2.6 and 5.2.2.7 shall be applied.

RE 2010-030E HMI_361 To permit to display each "Map attribute" area created when more than one predefined map is selected the following shall be displayed (see figure below):

- "Preview/Next" arrows (to move from a "Map attribute" area to the other)
- a counter (to indicate how much maps are composing the group)

5.2.4.4 Push buttons

5.2.4.4.1 "Defined Maps"

RE 2010-030E HMI_191 A click B1 on the "Defined Maps" push button shall allow to open the AIRSPACE MAPS cascading submenu where are located the predefined dynamic maps.

RE 2010-030E HMI_192 The cascading submenu of AIRSPACE MAPS shall allow to select a single or multiple predefined maps.

RE 2010-030E HMI_193 When a predefined map is selected in the cascading submenu of AISPACE MAPS the "autoclose" behaviour of the submenu on the display shall no more be possible.

RE 2010-030E HMI_194 When a predefined map is selected in the cascading submenu of AIRSPACE MAPS it shall be possible to close the submenu on the display with a click B2 on the title bar of the submenu.

RE 2010-030E HMI_195 A "Map attributes" area (see RE 2010-030E HMI_360) shall be created inside the "Dynamic Map Group Creation" window for each predefined map selected in the cascading submenu of AIRSPACE MAPS.

5.2.4.4.2 "Preview"

RE 2010-030E HMI_196 A click B1 on the "Preview" push button shall display temporarily the group of the selected maps.

5.2.4.4.3 "Add"

RE 2010-030E HMI_362 A click B1 on the "Add" push button when the focus is in a field inside the "Today/Repetitive Period" area shall add a new "Today/Repetitive Period" area (see RE 2010-030E HMI_153) and shall update the counter accordingly.

5.2.4.4.4 "Remove"

RE 2010-030E HMI_197 A click B1 on the "Remove" push button when the focus is in a field inside the "Map attributes" area shall remove the current creation settings (inputs are lost) and shall update the counter accordingly.

RE 2010-030E HMI_363 A click B1 on the "Remove" push button when the focus is in a field inside the "Today/Repetitive Period" area shall remove the current creation settings (inputs are lost) and shall update the counter accordingly.

5.2.4.4.5 "Send"

RE 2010-030E HMI_198 A click B1 on the "Send" push button shall close the "Dynamic Map Group Creation" window and the "Dynamic Airspace Creation" window.

RE 2010-030E HMI_199 When the "Dynamic Map Group Creation" window and the "Dynamic Airspace Creation" window is closed the inputs for the predefined map group shall be validated and shall be sent to SKYVISU server.

RE 2010-030E HMI_200 The SKYVISU server shall distribute the new dynamic map group to all the corresponding CWPs according to the sectors or units defined in § 5.2.2.4 and according the DZ.

RE 2010-030E HMI_201 If the created map is set to be activated on the current day an entry is added in the "Today Maps display" area in the "Dynamic Airspace management" window (RE 2010-030E HMI_002).

5.2.4.4.6 "Cancel"

RE 2010-030E HMI_202 A click B1 on the "Cancel" push button or a click B2 on the title bar of the window shall cancel the current creation settings (inputs are lost) and shall close the "Dynamic Map Group Creation" window and the "Dynamic Map Creation" window.

5.2.5 Dynamic Maps Creation – "DABS" cascading submenu

RE 2010-030E HMI_203 The "DABS" cascading submenu of "Maps" push button shall activate the following creation mode of dynamic maps:

- PIE/CIRCLE
- POINT LIST
- STRIP

RE 2010-030E HMI_204 The POLYGON creation mode shall be suppressed from the "DABS" cascading submenu.

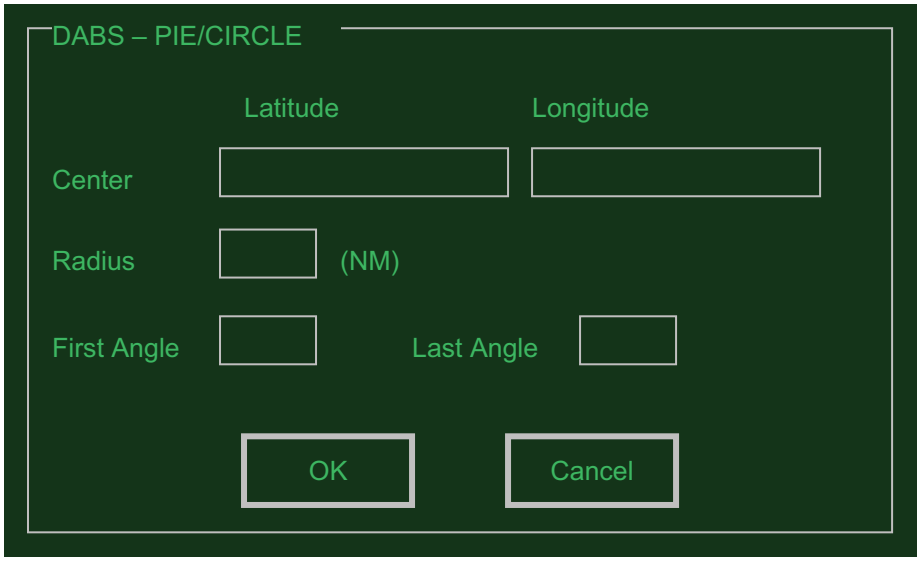
RE 2010-030E HMI_205 The CIRCLE creation mode shall be suppressed from the "DABS" cascading submenu.

5.2.5.1 PIE/CIRCLE

RE 2010-030E HMI_206 The PIE/CIRCLE creation mode shall be activated within a click B1 on the " PIE/CIRCLE " menu item of the "DABS" cascading submenu.

RE 2010-030E HMI_207 When the creation mode for PIE/CIRCLE is invoked the "DABS - PIE/CIRCLE" setting window shall be opened.

RE 2010-030E HMI_208 In the "DABS – PIE/CIRCLE" setting window editable text fields for Latitude, Longitude, Radius, First angle, Last angle and push buttons for OK and Cancel shall be available (see figure below).



RE 2010-030E HMI_209 A click B1 on the "OK" push button in the "DABS – PIE/CIRCLE" setting window shall validate the inputs, close the "DABS – PIE/CIRCLE" setting window and draw the

defined pie or circle.

RE 2010-030E HMI_210 When the defined pie or circle is drawn the indication "000/999" for lower and higher level limits shall appear under the mouse pointer.

RE 2010-030E HMI_211 When the indication "000/999" for lower and higher level limits appear under the mouse pointer, it shall be possible to move the "000/999" indication to the wanted position.

RE 2010-030E HMI_212 When the position of the "000/999" indication is selected a click B1 shall fix the "000/999" label as part of the map presentation and shall end the creation mode.

RE 2010-030E HMI_213 A click B1 on the "Cancel" push button in the "DABS – PIE/CIRCLE" setting window shall discard the inputs and close the window.

RE 2010-030E HMI_214 A click B2 on the title bar of the "DABS – PIE/CIRCLE" setting window shall discard the inputs and close the window.

RE 2010-030E HMI_215 The default values for the first angle and the last angle shall be respectively 0 and 360, which corresponds to a circle map.

RE 2010-030E HMI_216 If the user chooses a first angle value different form the 0 and/or a last angle value different from 360, it shall be create a Pie map instead of a circle map.

RE 2010-030E HMI_250 A click B1 on the "OK" push button in the "DABS – PIE/CIRCLE" setting window shall validate the inputs, close the "DABS – PIE/CIRCLE" setting window and open the DABS – Map attributes setting window to define the attributes of the defined pie or circle (see § 5.2.5.4).

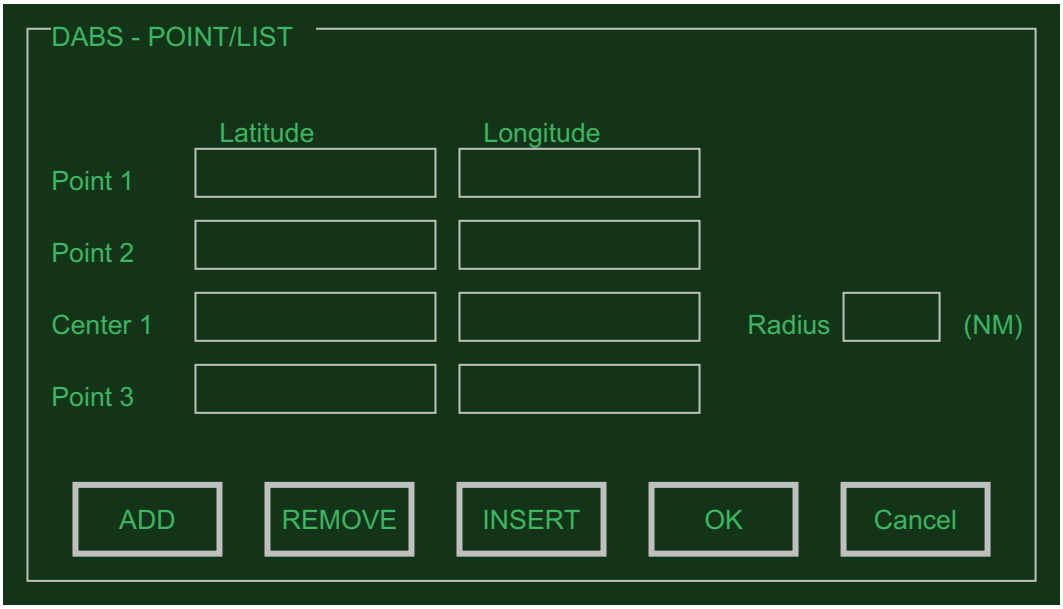
RE 2010-030E HMI_251 A click B1 on the "OK" push button in the "DABS – Map attributes" setting window shall validate the inputs, close the "DABS – Map attributes" setting window and draw the defined pie or circle.

5.2.5.2 POINT LIST

RE 2010-030E HMI_217 The POINT LIST creation mode shall be activated within a click B1 on the " POINT LIST " menu item of the "DABS" cascading submenu.

RE 2010-030E HMI_218 When the creation mode for POINT LIST is invoked the "DABS - POINT/LIST" setting window shall be opened.

RE 2010-030E HMI_219 In the "DABS – POINT/LIST" setting window editable text fields for Latitude and Longitude and push buttons for ADD, REMOVE, INSERT, OK and CANCEL shall be available (see figure below).



RE 2010-030E HMI_220 A click B1 on the "OK" push button in the "DABS – POINT/LIST" setting window shall validate the inputs, close the "DABS – POINT/LIST" setting window and draw the defined points.

RE 2010-030E HMI_221 When the defined points are drawn the indication "000/999" for lower and higher level limits shall appear under the mouse pointer.

RE 2010-030E HMI_222 When the indication "000/999" for lower and higher level limits appear under the mouse pointer, it shall be possible to move the "000/999" indication to the wanted position.

RE 2010-030E HMI_223 When the position of the "000/999" indication is selected a click B1 shall fix the "000/999" indication as part of the map presentation and shall end the creation mode.

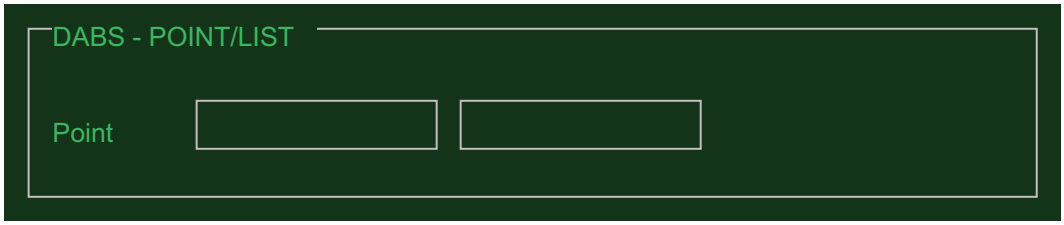
RE 2010-030E HMI_224 A click B1 on the "Cancel" push button in the "DABS – POINT/LIST" setting window shall discard the inputs and close the window.

RE 2010-030E HMI_225 A click B2 on the title bar of the "DABS – POINT/LIST" setting window shall discard the inputs and close the window.

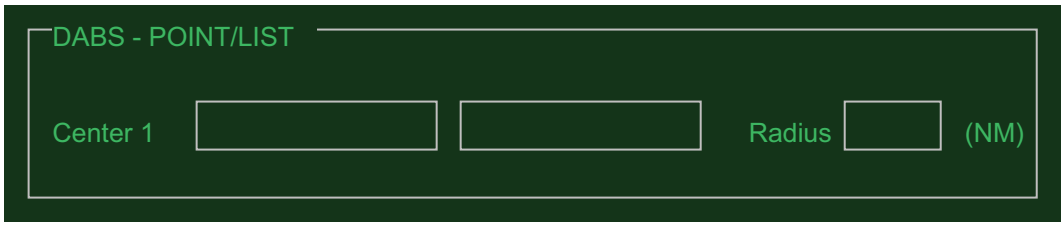
RE 2010-030E HMI_226 When the "DABS – POINT/LIST" setting window is opened, by default only one row for one point shall be proposed.

RE 2010-030E HMI_227 When the first row is completed and a click B1 is made on the "ADD" push button, the choice shall be given to add a supplementary row for a point or a supplementary row for an arc.

See below an example of a supplementary row for a point



See below an example for a supplementary row for an arc



RE 2010-030E HMI_228 When the focus is in a row in the "DABS – POINT/LIST" setting window a click B1 on the REMOVE push button shall remove the selected row.

RE 2010-030E HMI_229 When the focus is in a row in the "DABS – POINT/LIST" setting window a click B1 on the INSERT push button shall insert a new row before the selected one.

RE 2010-030E HMI_230 If the number of rows for the DABS – POINT/LIST Map exceeds the size of the "DABS – POINT/LIST" setting window a scroll bar shall be displayed on the left side of the window.

RE 2010-030E HMI_231 The system shall automatically close the polygon shape using the coordinates of the first entered point.

RE 2010-030E HMI_262 A click B1 on the "OK" push button in the "DABS – POINT/LIST" setting window shall validate the inputs, close the "DABS – POINT/LIST" setting window and open the DABS – Map attributes setting window to define the attributes of the defined points (see § 5.2.5.4).

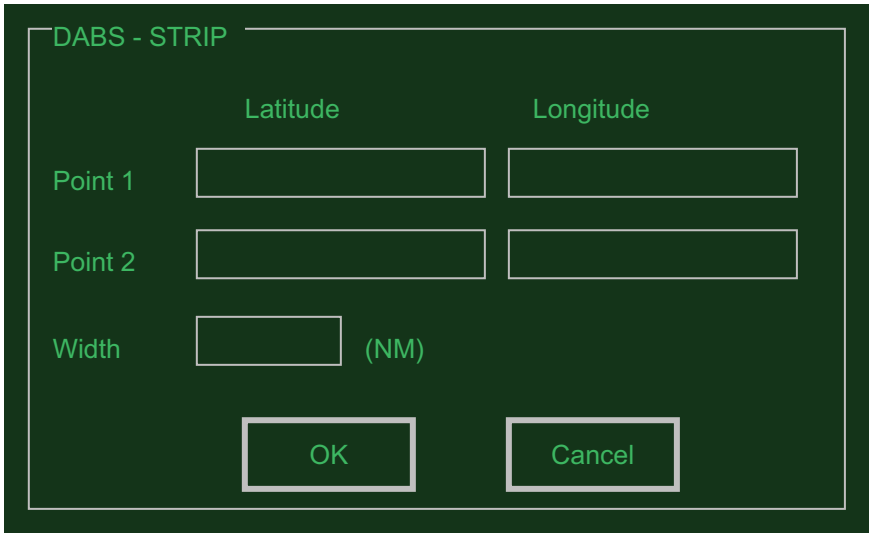
RE 2010-030E HMI_263 A click B1 on the "OK" push button in the "DABS – Map attributes" setting window shall validate the inputs, close the "DABS – Map attribute"s setting window and draw the defined points.

5.2.5.3 STRIP

RE 2010-030E HMI_232 The STRIP creation mode shall be activated within a click B1 on the " STRIP " menu item of the "DABS" cascading submenu.

RE 2010-030E HMI_233 When the creation mode for STRIP is invoked the "DABS – STRIP" setting window shall be opened.

RE 2010-030E HMI_234 In the "DABS – STRIP" setting window editable text fields for Latitude, Longitude and Width and push buttons for OK and CANCEL shall be available (see figure below).



RE 2010-030E HMI_235 The "Point1" "Latitude" and "Longitude" editable text fields available in the "DABS – STRIP" setting window shall allow to set the Latitude and Longitude coordinates of the starting point of the corridor.

RE 2010-030E HMI_236 The "Point2" "Latitude" and "Longitude" editable text fields available in the "DABS – STRIP" setting window shall allow to set the Latitude and Longitude coordinates of the ending point of the corridor.

RE 2010-030E HMI_237 The "Width" editable text field available in the "DABS – STRIP" setting window shall allow to set the width of the corridor in NM.

RE 2010-030E HMI_238 A click B1 on the "OK" push button in the "DABS – STRIP" setting window shall validate the inputs, close the "DABS – STRIP" setting window and draw the defined strip.

RE 2010-030E HMI_239 When the defined strip is drawn the indication "000/999" for lower and higher level limits shall appear under the mouse pointer.

RE 2010-030E HMI_240 When the indication "000/999" for lower and higher level limits appears under the mouse pointer, it shall be possible to move the "000/999" indication to the wanted position.

RE 2010-030E HMI_241 When the position of the "000/999" indication is selected a click B1 shall fix the "000/999" indication as part of the map presentation and shall end the creation mode.

RE 2010-030E HMI_242 A click B1 on the "Cancel" push button in the "DABS – STRIP" setting window shall discard the inputs and shall close the window.

RE 2010-030E HMI_243 A click B2 on the title bar of the "DABS – STRIP" setting window shall discard the inputs and shall close the window.

RE 2010-030E HMI_281 A click B1 on the "OK" push button in the "DABS – STRIP" setting window shall validate the inputs, close the "DABS – STRIP" setting window and open the "DABS – Map

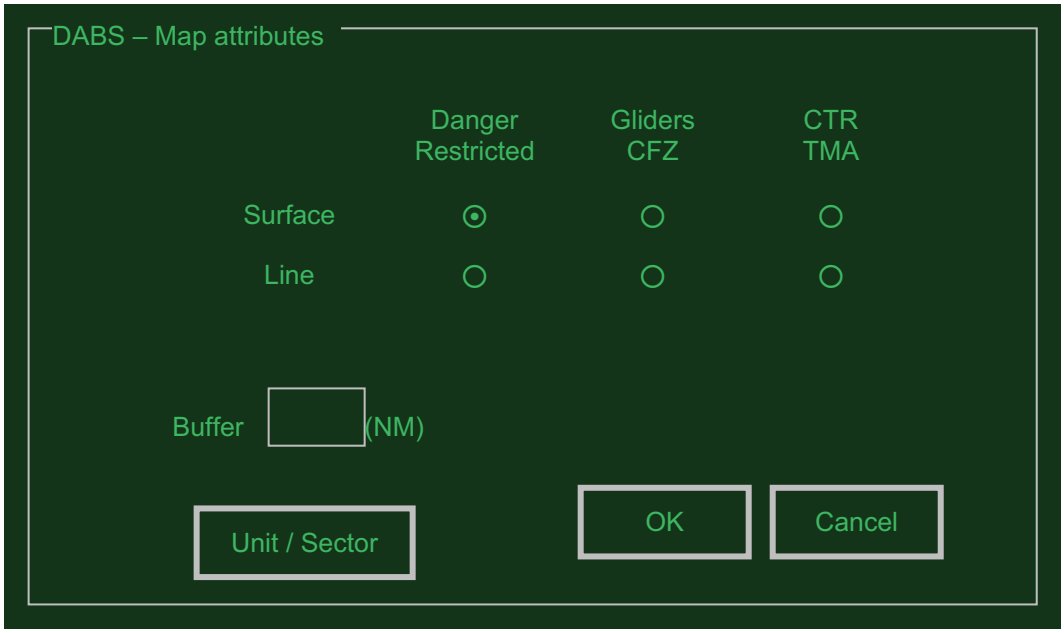
attributes" setting window to define the attributes of the defined points (see § 5.2.5.4).

RE 2010-030E HMI_282 A click B1 on the "OK" push button in the "DABS – Map attributes" setting window shall validate the inputs, close the "DABS – Map attribute"s setting window and draw the defined strip.

5.2.5.4 DABS – Map attributes setting window

RE 2010-030E HMI_288 The following shall be available In the "DABS – Map attributes" setting window (see the figure below as an example):

- option buttons for lines or surfaces for selectable attributes (Same attributes as for the predefined map shall be available)
- an editable field for the buffer value
- Push button "Unit / Sector" – see details in § 5.2.2.4



RE 2010-030E HMI_289 A click B1 on the "OK" push button in the "DABS – Map attributes" setting window shall validate the inputs, close the "DABS – Map attributes" setting window and draw the defined map at all the CWP corresponding to the sectors or units defined in § 5.2.2.4 and according the DZ.

RE 2010-030E HMI_290 When the defined map is drawn the shape of the map shall automatically be resized according the buffer value defined in the editable field in the "DABS – Map attributes" setting window.

RE 2010-030E HMI_291 When the defined map is drawn the type (colour and attribute) selected in the "DABS – Map attributes" setting window shall be automatically applied (same as the one used for predefined maps).

RE 2010-030E HMI_292 A click B1 on the "Cancel" push button in the "DABS – Map

attributes setting window shall discard the inputs and close the window.

5.2.6 Dynamic Maps Modification

RE 2010-030E HMI_293 A click B1 on the "Modify" push button when one dynamic map entry is selected shall open the "Dynamic Map Creation" window filled in with all the map information.

RE 2010-030E HMI_301 A click B1 on the "Modify" push button when a dynamic map entry of a group of maps (name of the group) is selected shall open the "Dynamic Map Group Creation" window filled in with all the information related to each map that composes the group.

RE 2010-030E HMI_294 A double click B1 on the selected dynamic map in the "Dynamic airspace window" ("Active Map", "Today Map" or "Repetitive Map") shall open the "Dynamic Map Creation" window filled in with all the map information.

RE 2010-030E HMI_303 A double click B1 on the selected dynamic map entry of a group of maps (name of the group) in the "Dynamic airspace window" ("Active Map", "Today Map" or "Repetitive Map") shall open the "Dynamic Map Group Creation" window filled in with all the information related to every single map that composes the group.

RE 2010-030E HMI_295 When the "Dynamic Map Creation" window is opened to modify a map the following map settings can be modified:

- Map start and ends days
- Map active days
- Map activation periods (start and end times)
- Map altitude range
- The altitude range level correction check boxes status
- The text comments
- The display at unit and sector

RE 2010-030E HMI_305 When the "Dynamic Map Group Creation" window is opened to modify a group of map it shall be allowed to modify the following map setting for all the maps in the group:

- Map start and ends days
- Map active days
- Map activation periods (start and end times)

RE 2010-030E HMI_306 When the "Dynamic Map Group Creation" window is opened to modify a group of map it shall be allowed to modify the following map setting for every single map that composes the group:

- Map altitude range
- The altitude range level correction check boxes status
- The text comments
- The display at unit and sector

RE 2010-030E HMI_296 Once the modification is performed a click B1 on the "Send" push button of the "Dynamic Map Creation" window shall close the window, shall validate the modifications shall send the created map to the SKYVISU server.

RE 2010-030E HMI_308 Once the modification is performed in the "Dynamic Map Group Creation" window a click B1 on the "Send" push button shall close the window, shall validate the modifications shall send the created map group to the SKYVISU server.

RE 2010-030E HMI_297 The modified map shall be distribute to all the corresponding CWPs.

RE 2010-030E HMI_298 If the modified map is set to be activated on the current day an entry is added in the "Today Maps" display area in the "Dynamic Airspace management" window (RE 2010-030E HMI_002).

RE 2010-030E HMI_299 At any moment, a click B1 on the "Cancel" push button of the "Dynamic Map Creation" window shall cancel the current modifications and shall close the window.

5.2.7 Dynamic Maps Deletion

RE 2010-030E HMI_312 A click B1 on the "Delete" push button when one dynamic map entry is selected shall delete the map.

RE 2010-030E HMI_319 A click B1 on the "Delete" push button when a dynamic map entry of a group of dynamic map (name of the group) is selected shall delete all the map that composes the corresponding group of dynamic map.

RE 2010-030E HMI_313 An acknowledge popup window shall be displayed to confirm the deletion.

RE 2010-030E HMI_314 When the "Ok" push button of the confirmation window is pressed and the selected map for deletion is located in the "Active Maps" area, the map shall be removed from display and the map entry in the DAM window shall be moved to the "Today Maps" area with the "Terminated" indication.

RE 2010-030E HMI_322 When the "Ok" push button of the confirmation window is pressed and the selected group of map (name of the group) for deletion is located in the "Active Maps" area, the group of map shall be removed from display and the map entry in the DAM window shall be moved to the "Today Maps" area with the "Terminated" indication.

RE 2010-030E HMI_315 When the "Ok" push button of the confirmation window is pressed and he selected map for deletion is located in the "Today Maps" area, the map entry shall remains on the "Today Maps" area with the "Terminated" indication. If several activation periods are available for the same map instance, only the selected period shall be cancelled. If several activation periods for one same map are cancelled, only one entry with the "Terminated" indication shall be kept

in the "Today Maps" area.

RE 2010-030E HMI_324 When the "Ok" push button of the confirmation window is pressed and the selected group of map (name of the group) for deletion is located in the "Today Maps" area, the map entry shall remains on the "Today Maps" area with the "Terminated" indication. If several activation periods are available for the same group of map instance, only the selected period shall be cancelled. If several activation periods for one same group of map are cancelled, only one entry with the "Terminated" indication shall be kept in the "Today Maps" area.

RE 2010-030E HMI_316 When the "Ok" push button of the confirmation window is pressed and the selected map for deletion is located in the "Repetitive/Planned Maps" area, the map entry shall be removed from the list but if the repetitive map was planned to be active on the current day the current day entry or entries (in the "Active Maps" or "Today Maps" areas) shall be remain unchanged.

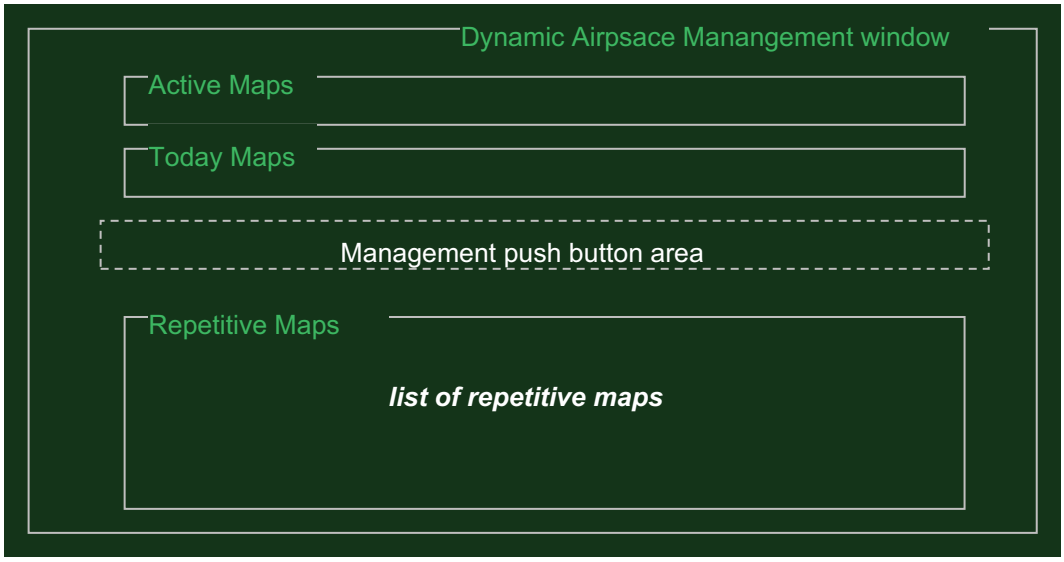
RE 2010-030E HMI_326 When the "Ok" push button of the confirmation window is pressed and the selected group of map (name of the group) for deletion is located in the "Repetitive/Planned Maps" area, the map entry shall be removed from the list but if the repetitive map was planned to be active on the current day the current day entry or entries (in the "Active Maps" or "Today Maps" areas) shall be remain unchanged.

RE 2010-030E HMI_317 When the "Ok" push button of the confirmation window is pressed and the selected map for deletion is a repetitive one, the map indication and all its instances, if any, located in the "Active Maps" or "Today Maps" lists shall be removed.

RE 2010-030E HMI_328 When the "Ok" push button of the confirmation window is pressed and the selected group of map (name of the group) for deletion is a repetitive one, the map indication and all its instances, if any, located in the "Active Maps" or "Today Maps" lists shall be removed.

5.2.8 Repetitive Dynamic Maps List

RE 2010-030E HMI_329 A click B1 on the "Rep. Map" toggle button shall extend the DAM window to display a Repetitive Dynamic Map List (part on the bottom in the figure below).



RE 2010-030E HMI_330 The Repetitive Dynamic Map List shall consist in a List similar to the "Active Maps" and "Today Maps" lists but introduced with the label Repetitive/Planned Maps"

RE 2010-030E HMI_331 For map entries within the "Repetitive Dynamic Maps" list the following information shall be available:

- Active period from start day to end day
- Active days within the activation period ("M" for Monday, "T" for Tuesday, …
  If the map is not active for one day in the activation period, the corresponding letter is replaced by a dash "-"
- Map name

RE 2010-030E HMI_332 The text related to the "Repetitive Maps" shall be displayed entirely up to the frame of the available space in the corresponding window (same as described in RE 2010-030E HMI_004).

ⓘ Shall not be truncated as it is the case today

ⓘ The size of the window shall not change

RE 2010-030E HMI_333 If the text related to the "Repetitive Maps" exceeds the frame of the text delimitation (window) a hook on the line shall permit to display the text entirely (same as described in RE 2010-030E HMI_005).

5.2.9 Dynamic Map Information Display

RE 2010-030E HMI_335 The dynamic map activated by FMP operator shall be displayed at all the CWP according to the corresponding sector or unit (see § 5.2.2.4) and according to the DZ.

RE 2010-030E HMI_336 The dynamic map shall be represented by a full surface or by a border line.

5.2.9.1 First usable level / Start – End time

RE 2010-030E HMI_337 The lower and upper level indication of the dynamic map activated by FMP operator shall be displayed at all the CWP according to the corresponding sector or unit defined in § 5.2.2.4 and according to the DZ.

RE 2010-030E HMI_338 If displayed, it shall be possible to move the lower and upper level indication of an activated dynamic map to the wanted position (drag and drop) inside the shape of the activated dynamic map.

RE 2010-030E HMI_339 If the DZ of the CWP is not totally included in the map altitude range the lower and upper level indication of the activated dynamic map shall be performed as described in the following tables:

In Geneva:

Map Levels		Level Indication	
Low Level	High Level	TSA Maps	Non TSA Maps
000	YYY	YYY(+FL Correction if ticked)+	
XXX	YYY	XXX(+FL Correction if ticked)-/ YYY(+FL Correction if ticked)+	
XXX	>=630	XXX-	
000	>=630	No Display	

In Zurich:

Map Levels		Level Indication	
Low Level	High Level	TSA Maps	Non TSA Maps
000	YYY	-YYY	YYY+Correction(1)
XXX	YYY	XXX/YYY	YYY+Correction(1) -XXX
XXX	630	XXX+	-XXX
000	630	No Display	

(1) This correction depends on the high level

RE 2010-030E HMI_340 If the DZ of the CWP is fully included in the map altitude range, the lower and upper level indication of the activated dynamic map shall not be displayed.

RE 2010-030E HMI_361 In case of TSA map activation, if the DZ of the CWP is not totally included in the map altitude range, the lower and upper level indication of the activated dynamic map shall be performed to the CWP.

ⓘ GVA only

RE 2010-030E HMI_341 If the Start Time check box has been ticked in the "Dynamic Map Creation" window (see § 5.2.2.3), the remaining time before activation (with the format: "S:<MM>", standing for: Start Time: Minutes) shall be displayed in the map label 15 minutes before its activation.

ⓘ No activation indication shall be displayed for TSA

maps (Only ZRH)

RE 2010-030E HMI_342 If the End Time check box has been ticked in the "Dynamic Map Creation" window (see § 5.2.2.3), the remaining time before de-activation (with the format: "E:<MM>", standing for End Time: Minutes) shall be displayed in the map label if the map will be de-activated in less than 15 minutes.

RE 2010-030E HMI_343 The indication of the remaining time shall be displayed in white colour.

RE 2010-030E HMI_344 In case of subsequent activation of the same dynamic map with different lower or upper level, both the "End" and the "Start" counter with the corresponding FL shall be displayed:

E:15 100+
S:15 300+

ⓘ A click B3 on the area displays all activities related to the dynamic map.

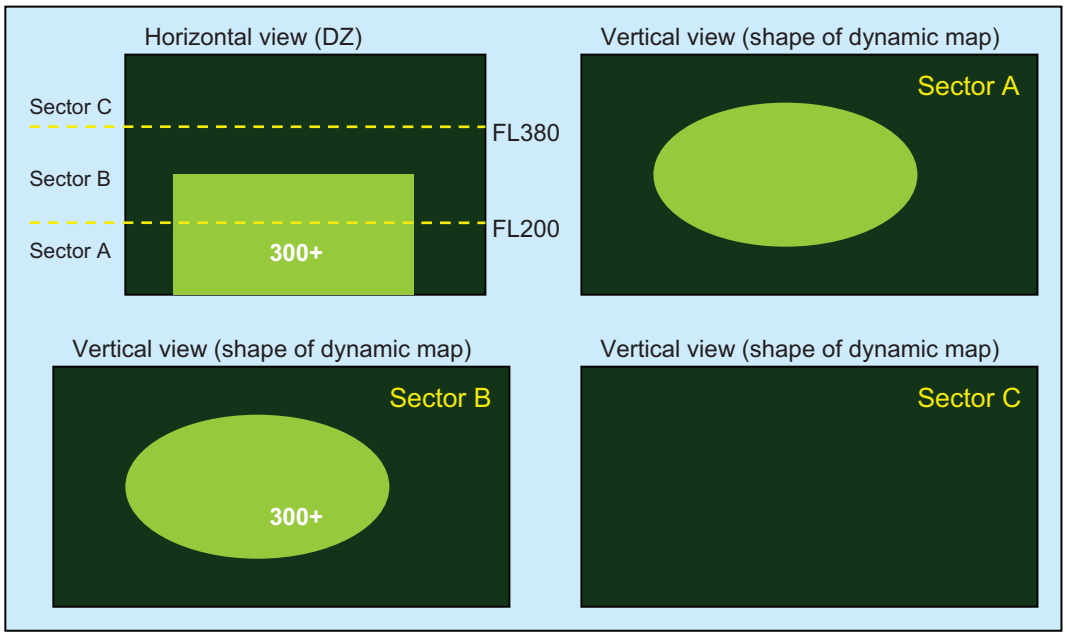
RE 2010-030E HMI_345 If the "Display Levels" check box is checked the FL (first usable level) shall be displayed as described in the table above (see RE 2010-030E HMI_339).

5.2.9.2 Dynamic map

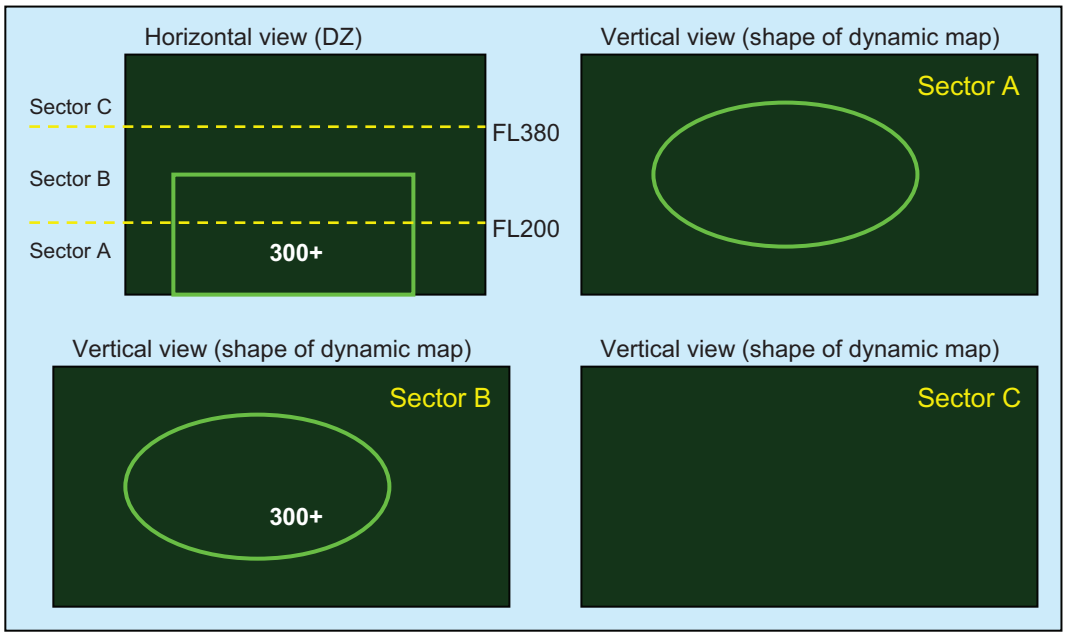
RE 2010-030E HMI_346 All the dynamic map activated by FMP operator shall be displayed at all the CWP corresponding to the sectors or units defined in § 5.2.2.4 and according to the DZ.

ⓘ The figures below are an example: according to the DZ ⇨ the dynamic map shall be displayed for the sector A and B and not for the Sector C.

Dynamic map represented by a **full surface**:



Dynamic map represented by a **border line**:

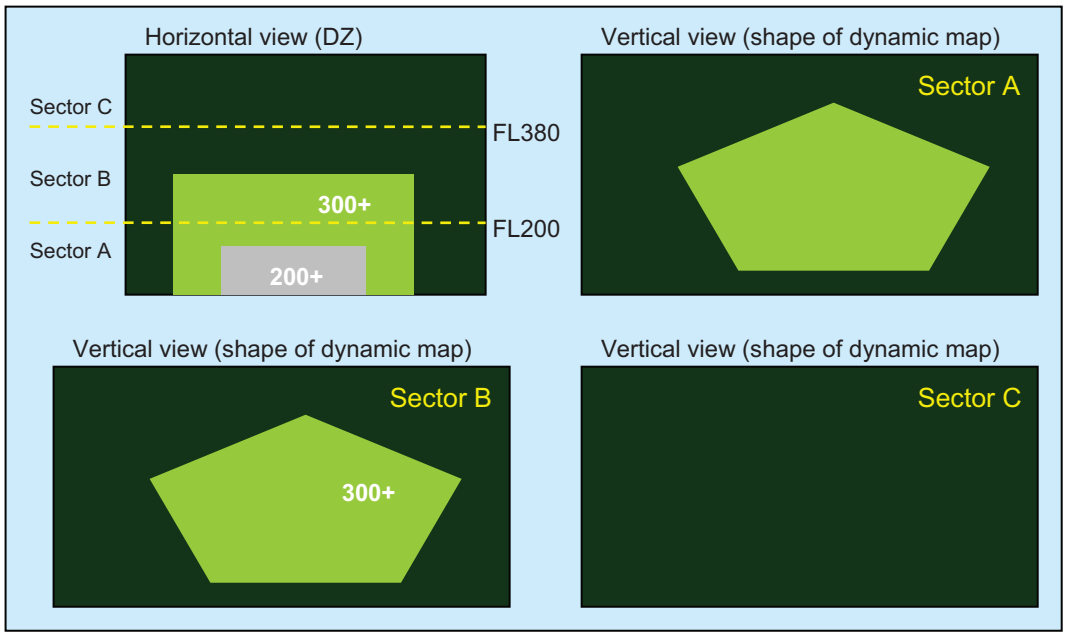


RE 2010-030E HMI_347

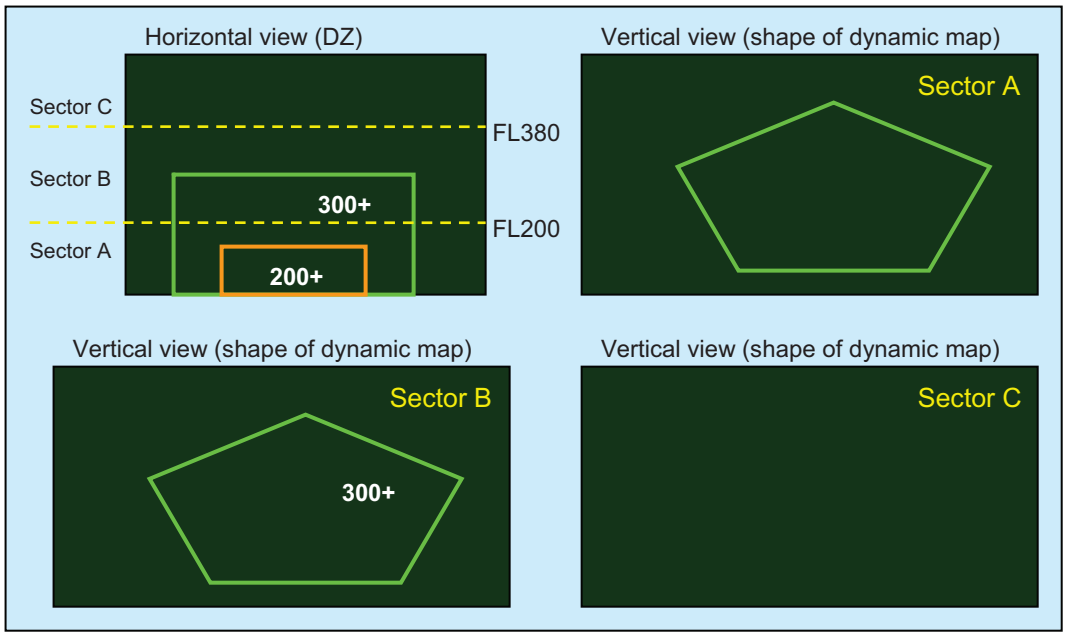
If a dynamic map overlap entirely another one, the volume of the biggest one shall be displayed.

ⓘ The figures below are an example:

Dynamic map represented by a **full surface**: the grey area shall not be displayed



Dynamic map represented by a **border line**: the orange line shall not be displayed

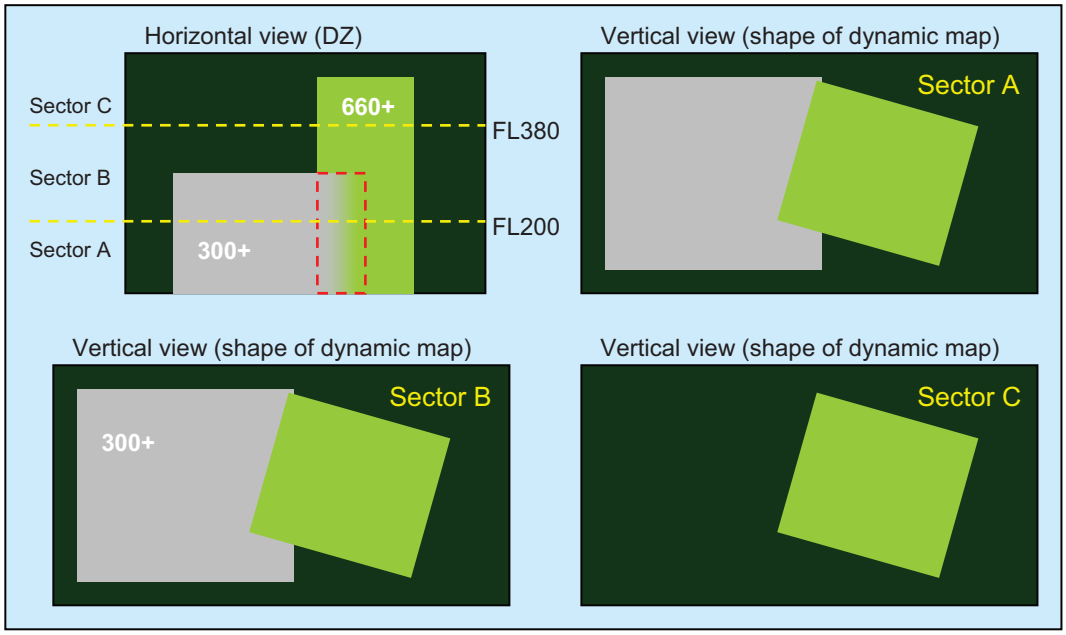


RE 2010-030E HMI_348

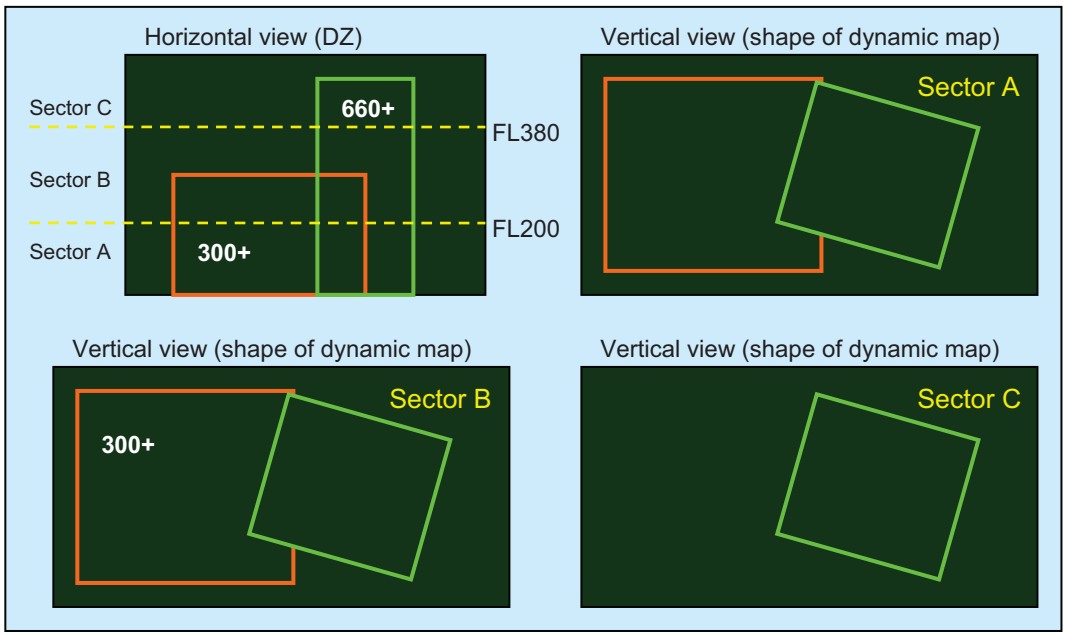
If a dynamic map overlap partially another one, the most penalizing volume shall be displayed.

ⓘ The figures below are an example:

Dynamic map represented by a **full surface**: the overlapped part of the grey area shall not be displayed.



Dynamic map represented by a **border line**: the overlapped part of the orange line shall not be displayed.

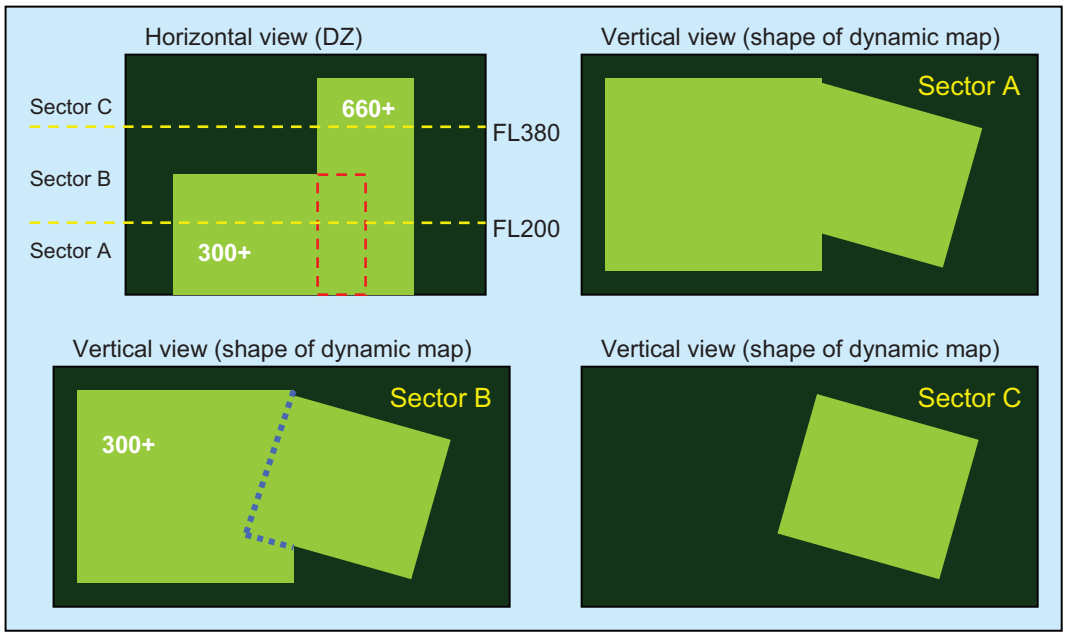


RE 2010-030E HMI_349

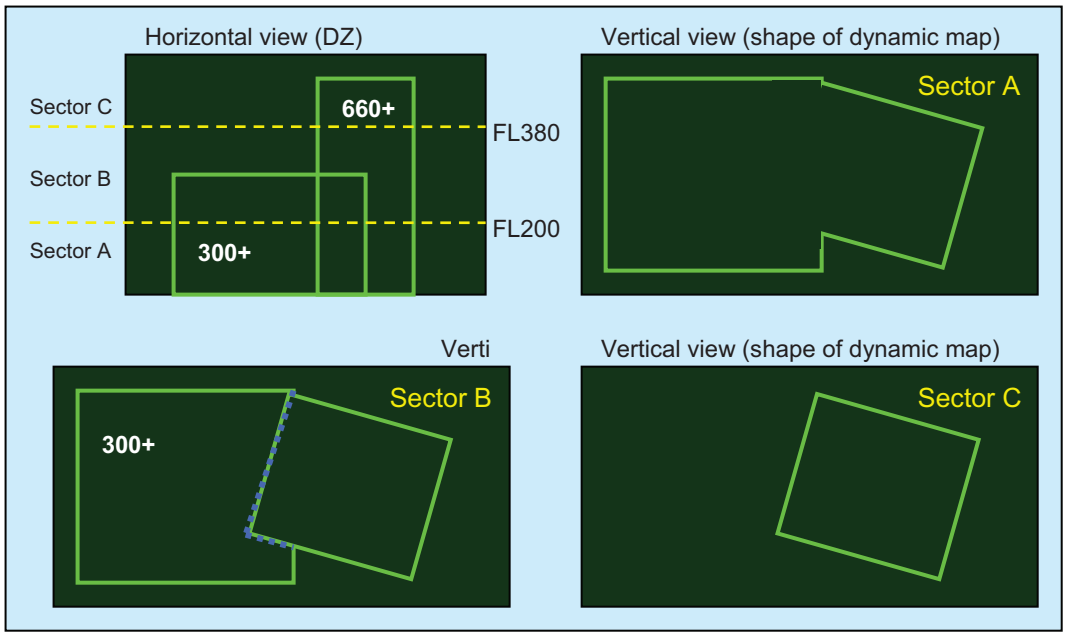
If the overlapping dynamic map belong to the same category of dynamic map a dashed line shall be displayed to indicate the limits of the highest shape in the zone of overlapping (Sector B).

ⓘ The figures below are an example:

Dynamic map represented by a **full surface**:



Dynamic map represented by a **border line**:



RE 2010-030E HMI_350

The behaviour described in RE 2010-030E HMI_349is not valid if the maps are entirely included in the sector according the DZ (Sector A).

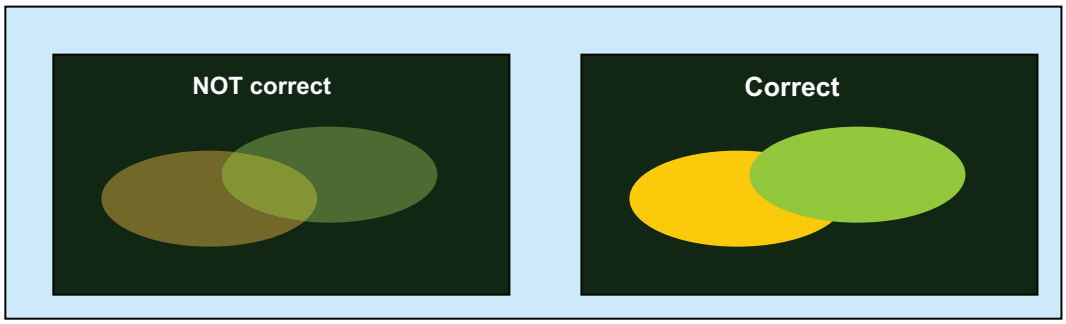
ⓘ The figures above are an example.

5.2.9.3 Transparency

RE 2010-030E HMI_353

If a dynamic map overlap partially another dynamic map and both dynamic map are displayed with a surface, it shall not create a third surface.

ⓘ The figure below is an example:



5.3 Parameters Requirements

RE 2010-030E PAR_001 Single predefined dynamic map shall be created for each TSA, glider activities (permanent glider zones, temporary glider zones, cloud flying zones), anti hail firing zones, avalanches firing zones, danger zones and parachute zones.

RE 2010-030E PAR_002 Each predefined map shall have following optional parameters:

- Low level (displayed/not displayed) § 5.3.1
- High level (displayed/not displayed) § 5.3.1
- FL or feet § 5.3.1
- Display Level § 5.3.2
- Start time of the activity § 5.3.3
- End time of the activity § 5.3.3
- Unit to display § 5.3.4
- Sector to display § 5.3.4
- Altitude corrections § 5.3.5

ⓘ Examples how it is currently defined:
LEVEL/ll=000/hl=330/tsa/bt/et
LEVEL/ll=000/ft/hl=7000/bt/et
LEVEL/ll=000/hl=999

RE 2010-030E PAR_003 It shall be possible to restore all the DAM maps ATCOs settings made manually from the FMP in case of SKYVISU server failure.

5.3.1 Low level / High level, flight level or feet

RE 2010-030E PAR_004 It shall be possible to define "Low level", "High level" as a parameter of the predefined dynamic map: "ll" standing for "Low level" and "hl" standing for "High level".

RE 2010-030E PAR_006 The value defined for the "Low level" shall be performed into "Low level" text field (see RE 2010-030E HMI_032).

RE 2010-030E PAR_007 The value defined for the "High level" shall be performed into "Higher" level text field (see RE 2010-030E HMI_032)

RE 2010-030E PAR_005 It shall be possible to define as a parameter of the

predefined dynamic map if the value of "Low level" and of "High level" is in "feet" or not: "ft" standing for "feet" (see RE 2010-030E HMI_034 and RE 2010-030E HMI_035).

RE 2010-030E PAR_008 The behaviours of the "ft" parameter shall be:

- If "ft" is defined, the check box associated to "Feet" in the Dynamic Airspace Management window is ticked
- If "ft" is not defined, the check box associated to "Feet" in the Dynamic Airspace Management window is un-ticked

5.3.2 Display Level

RE 2010-030E PAR_009 It shall be possible to define the display/no display of the "Low level" or of the "High level" at CWP as a parameter of the predefined dynamic map: "dl" is standing for "Display Level".

ⓘ See also RE 2010-030E HMI_045

RE 2010-030E PAR_010 The behaviours of the "Display Level" parameter shall be:

- If "dl" is defined, the check box associated to "Display Level" in the Dynamic Airspace Management window is ticked
- If "dl" is not defined, the check box associated to "Display Level" in the Dynamic Airspace Management window is un-ticked

ⓘ See also RE 2010-030E HMI_339

RE 2010-030E PAR_011 It shall be possible to define the size of the Font and the colour of the "Display Level" parameter.

5.3.3 "Start" and "End" time of activity

RE 2010-030E PAR_012 It shall be possible to define "Start time" or "End time" as a parameter of the predefined dynamic map: "bt" is standing for "Start time" and "et" is standing for "End time".

RE 2010-030E PAR_013 The behaviours of the "Start" parameter shall be:

- If "bt" is defined, the check box associated to "Start" time in the Dynamic Airspace Management window is ticked
- If "bt" is not defined, the check box associated to "Start" time in the Dynamic Airspace Management window is un-ticked

RE 2010-030E PAR_014 The behaviours of the "End" parameter shall be:

- If "et" is defined, the check box associated to "End" time in the Dynamic Airspace Management window is ticked.
- If "et" is not defined, the check box associated to "End" time in the Dynamic Airspace Management window is un-ticked.

5.3.4 Unit / Sector

RE 2010-030E PAR_015 It shall be possible to define "Unit" or "Sector" as a parameter of the predefined dynamic map.

RE 2010-030E PAR_016 The following "Unit" (column 2) and "Sector" (column 3) abbreviation shall used as a parameter of the predefined dynamic map:

Geneva			
Physical sector	Name of Unit in the MAP parameter	Name of Sector in the MAP parameter	Name of the check box in the "Display Position" window[1]
SECTOR1	LAC	INN	INN
SECTOR2	LAC	INE	INE
SECTOR3	LAC	INS	INS
--	LAC	INL	INL
SECTOR4	UAC	UL1	L1
SECTOR5	UAC	UL2	L2
SECTOR6	UAC	UL3	L3
SECTOR7	UAC	UL4	L4
SECTOR8	UAC	UL5	L5
SECTOR9	UAC	UL6	L6
SECTOR10	FIC	FIC	FIC
SECTOR11	FIC	DLT	DLT
SECTOR12	FIC	MM1	MIL
SECTOR13	FIC	MM2	MIL
SECTOR24	APP	DEP	DEP
SECTOR25	APP	ARR	ARR
SECTOR26	APP	FIN	FIN
SECTOR27	APP	PRN	PRN
--	SPV	SPV	SPVR
--	SPV	FMP	FMP

[1] see § 5.2.2.4

Zurich			
Physical sector	Name of Unit in the MAP parameter	Name of Sector in the MAP parameter	Name of the check box in the "Display Position" window[1]
--	LAC	LOW	W
--	LAC	LOS	S
--	LAC	LOE	E
--	LAC	LON	N
--	UAC	UM1	M1
--	UAC	UM2	M2
--	UAC	UM3	M3
--	UAC	UM4	M4
--	UAC	UM5	M5
--	UAC	UM6	M6

--	FIC	FIC	FIC
--	FIC	DLT	DLT
--	FIC	ARF	ARFA
--	APP	APW	APW
--	APP	FIN	FIN
--	APP	APE	APE
--	APP	CAP	CAP
--	APP	DEP	DEP
--	APP	RSV	RSV
--	APP	PRN	PRN
--	SPV	SPV	SPVR
--	SPV	FMP	FMP

[1] see § 5.2.2.4

RE 2010-030E PAR_017

The behaviours of the "Unit" parameter shall be:

- If a "Unit" is defined, the check box associated to the corresponding "Unit" in the Position to display" window is ticked.
- If a "Unit" is not defined, the check box associated to the corresponding "Unit" in the Position to display" window is un-ticked.

ⓘ See also § 5.2.2.4

RE 2010-030E PAR_018

The behaviours of the "Sector" parameter shall be:

- If a "Sector" is defined, the check box associated to the corresponding "Sector" in the Position to display" window is ticked.
- If a "Sector" is not defined, the check box associated to the corresponding "Sector" in the Position to display" window is un-ticked.

ⓘ see more § 5.2.2.4

5.3.5 Altitude corrections

RE 2010-030E PAR_023

It shall be possible to define "Altitude correction" as a parameter of the predefined dynamic map: "flc" is standing for "FL corr" and "Alps", "qnh" is standing for "QNH corr".

RE 2010-030E PAR_024

The behaviours of the "Altitude correction" for "FL corr"/"Alps" parameters shall be:

- If "flc" is defined, the check box associated to "FL corr"/"Alps" in the Dynamic Airspace Management window is ticked.
- If "flc" is not defined, the check box associated to "FL corr"/"Alps" in the Dynamic Airspace Management window is un-ticked.

RE 2010-030E PAR_025

The behaviours of the "Altitude correction" for "QNH corr" parameters shall be:

- If "qnh" is defined, the check box associated to "QNH corr" in the Dynamic Airspace Management window is ticked.
- If "qnh" is not defined, the check box associated to "QNH corr" in the Dynamic Airspace Management window is un-ticked.

Appendix 1 Distribution List

To :	Name	Service	Draft	Proposed	Released
	BACKER Freddy	TDA		X	X
	BARRAS Jacques	TDS		X	X
	BERGER Manuela	OOTE		X	X
	BOSONNET Lucien	OORB	X	X	X
	BUEHLER Mike	TDZ			X
	CACHIN Philippe	TDZ			X
	COLLINS Arlette	OZDV	X	X	X
	COMMINOT Stephan	OGCE	X	X	X
	CORBOZ Jérôme	OGDS		X	X
	DI PALMA Claudio	OZ-S			X
	DOSTER Thomas	OZC-T	X	X	X
	FANKHAUSER René	OMLZ	X	X	X
	GARRIGUES Benoît	TDA		X	X
	GEIGER Patrick	TDA			X
	GRIVET Michel	TDG			X
	JAVET Didier	OOTP	X	X	X
	JOMINI Cédric	OMLG	X	X	X
	JOSSEN Roger	OZDO	X	X	X
	MAEDER Michael	OORC		X	X
	MOTTA Andrea	TDA			X
	NIEDERER Peter	OZT-T	X	X	X
	OPS Safety request	SAO	X	X	X
	PERONE Dominique	OGCA	X	X	X
	PFEIFFER Manfred	OORC	X	X	X
	POURRAZ Fabrice	TDA			X
	REY Christian	TDA			X
	ROCHAT Ivan	OG-S		X	X
	SEILER Philipp	OZC-P	X	X	X
	STRAUMANN Matthias	OZT-P	X	X	X
	TABTI Saïd	OORB		X	X
	VILLIGER Tobias	OORC	X	X	X
	VOGEL Vanni	OGTA	X	X	X
	ZOULALIAN Alexandre	OGTE	X	X	X

Cc :	Name	Service	Draft	Proposed	Released
	BAUMANN Patrick	OZC		X	X
	HEINZER Xaver	OZ		X	X
	HOCHSTRASSER Pascal	OGT		X	X
	KIRTZ Pascale	OZT		X	X
	LANDHERR Fabian	OML		X	X
	LORI Beatrix	OZD		X	X
	MICHEL Raphaël	OOR		X	X
	PEREZ Julian	OGD		X	X
	PRADEL Johann	OGC		X	X
	SOMMER Peter	OG		X	X